POLITECNICO DI MILANO

Biochip Course

Microfabrication

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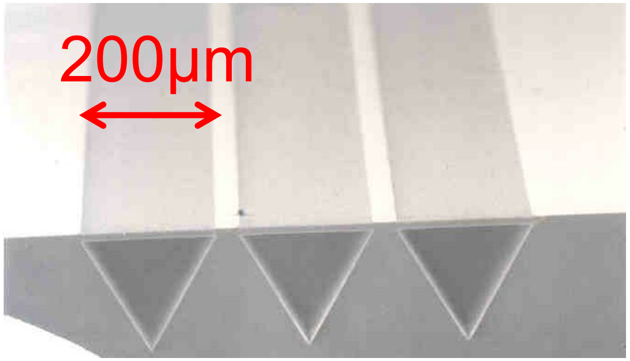
Biochips: When Microfabrication is Needed

To realize:

Microfluidic channels

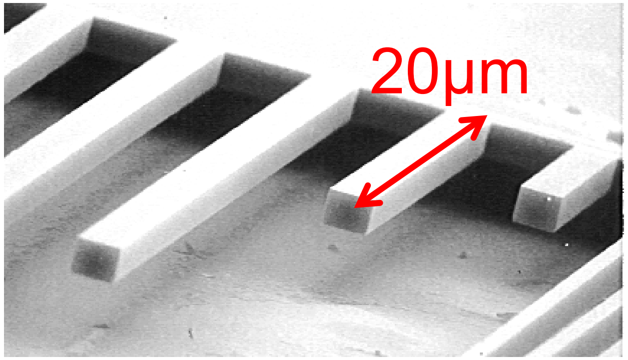
(in silicon, glass or elastomers)



Cantilevers or membranes

(oscillanting MEMS microbalances)

mechanical detection



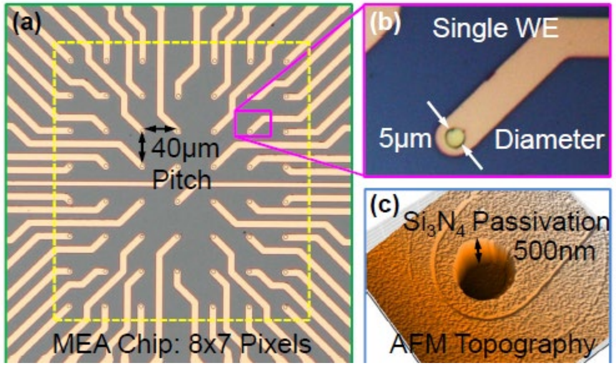
Integrated light waveguides

optical detection

Planar metal microelectrodes

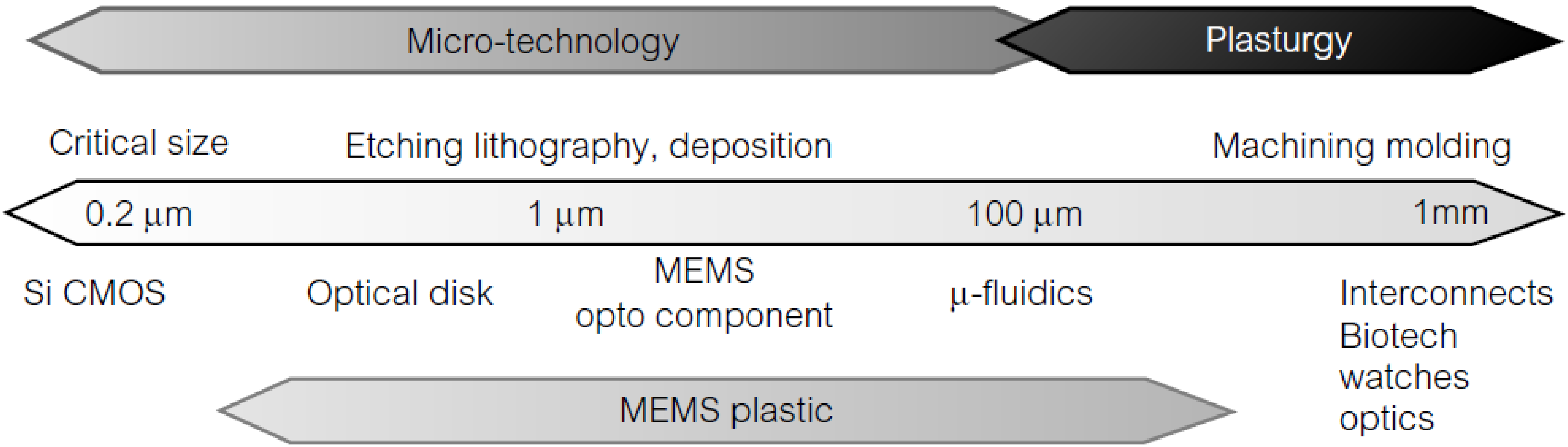
(in gold, platinum)

electrochemical detection



Fabrication Approaches

Selection of the proper technology that match the structure size:



- Additive (3D printers) vs subtractive (milling) machining
- Several materials are available
- Hard (silicon) vs. soft (plastic) technologies

Hard vs. Soft Fabrication Technologies

Hard	Soft
• 0.2 – 500 µm	• 10 µm - 10 mm
• Clean-room based	• Lab bench-top
• Expensive	• Low cost
• Silicon, glass	• Elastomers (PDMS) and plastics (PMMA)
• Starts from wafer	• Starts from liquid polymer
• Etching	• Replica molding
• Lithography (mask)	
• Deposition	

Mold

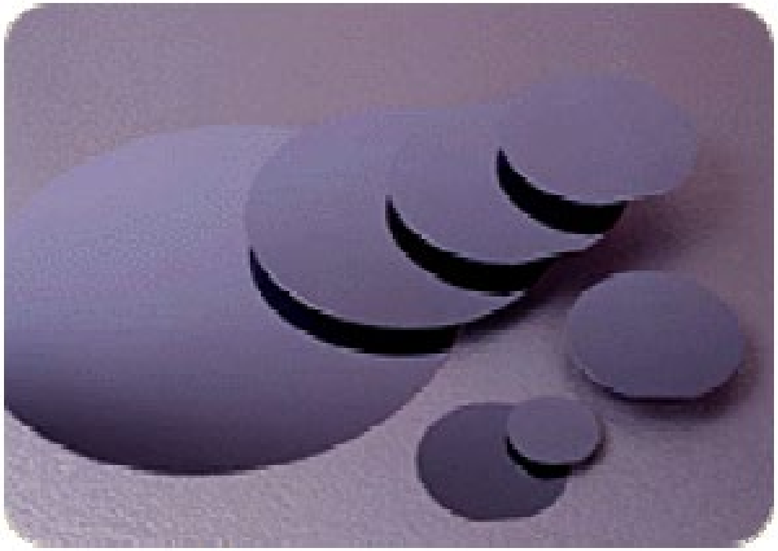
The Clean Room

Dedicated facility with air control and filtering to reduce the concentration of potentially dangerous micrometric airborne dust

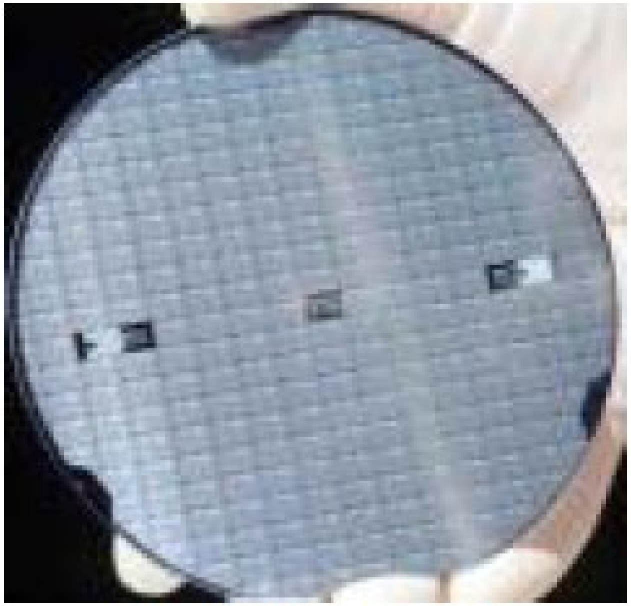
PoliFab: the new 400m^2 clean room @PoliMi

Clean Room Class	Maximum Particles per Cubic Foot (ft^3)					ISO Equivalent
	≥ 0.1 µm	≥ 0.2 µm	≥ 0.3 µm	≥ 0.5 µm	≥ 5.0 µm	
1	30	7	3	1		ISO 3
10	350	75	30	10		ISO 4
100		750	300	100		ISO 5
1,000				1,000	7	ISO 6
10,000				10,000	70	ISO 7
100,000				100,000	700	ISO 8

Preferred Substrate: Silicon Wafers



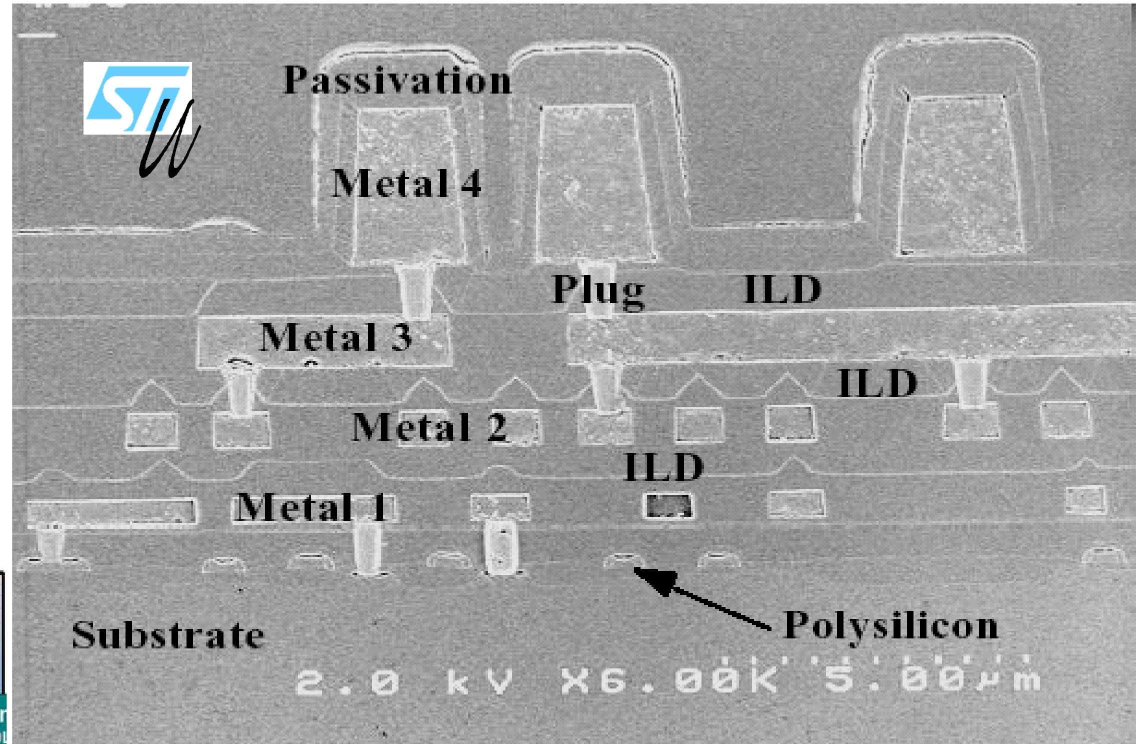
www.addisonengineering.com

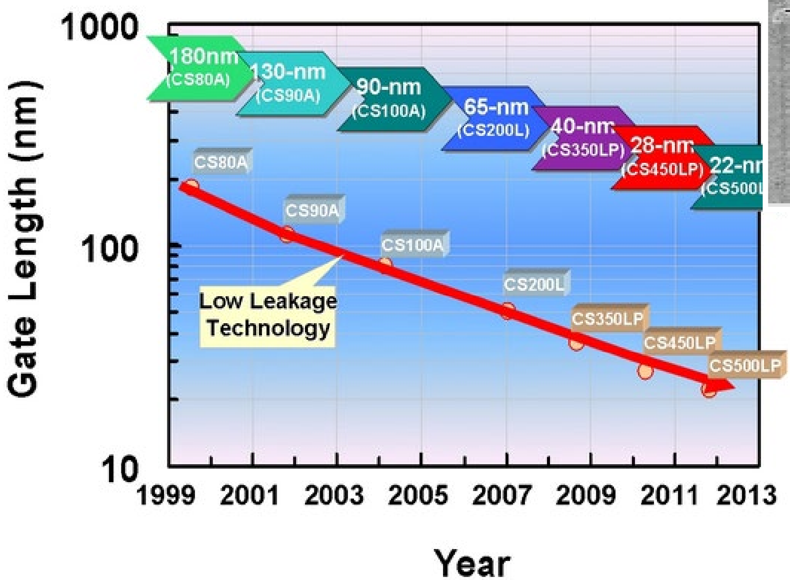


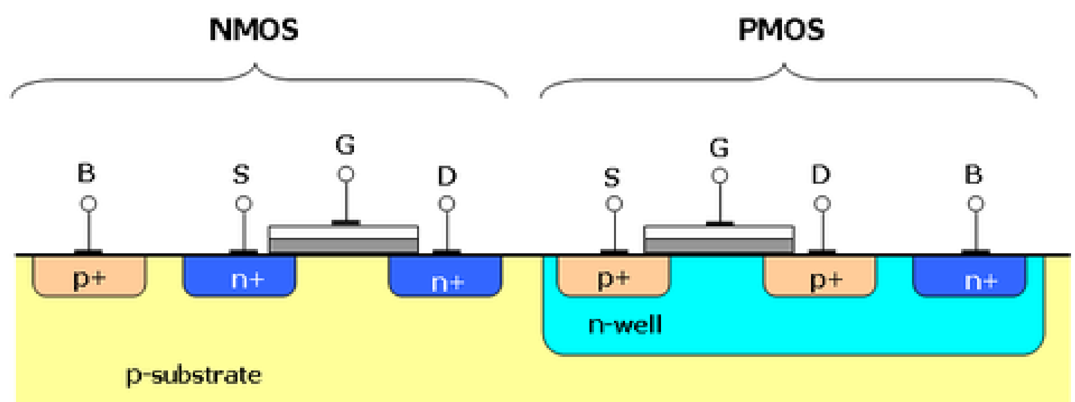
engr.calvin.edu

- Monocrystalline Si (500µm thick)
- Different doping levels and diameters (2” – 12”)
- Other materials available: SOI, glass and quartz (SiO_2)

The Standard CMOS Process







http://www.fujitsu.com/

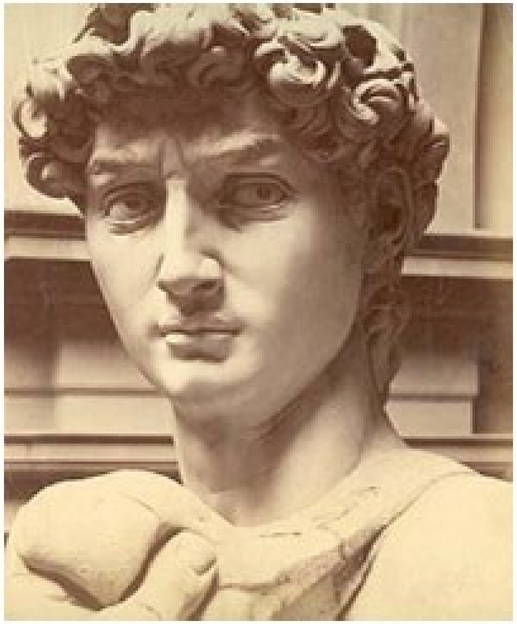
Silicon as a Building Material (MEMS)

Not only a substrate for transistors, also a structural material

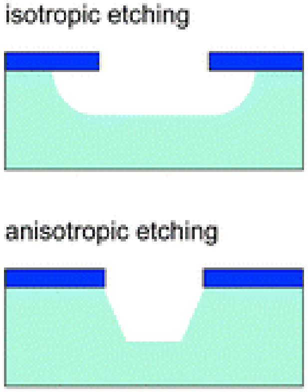
	Si	C	SiC	SiN	Fe	W	Steel	Mo	Al
Yield Strength (Gpa)	7	53	21	14	12.6	4	2.1	2.1	0.17
Knoop Hardness (Kg/mm^2)	850	7000	2480	3486	400	485	660	275	130
Young Modulus (100 Gpa)	1.9	10.3	7	3.8	1.96	4.1	2	3.43	0.7
Density (g/cm^3)	2.3	3.5	3.2	3.1	7.8	19.3	7.9	10.3	2.7
Thermal Conductivity (W/cm K)	1.57	20	3.5	0.19	0.8	1.78	0.32	1.38	2.36
Thermal Expansion (ppm/K)	2.33	1	3.3	0.8	12	4.5	17.3	5	25

MEMS Fabrication Approaches

Bulk Micromachining

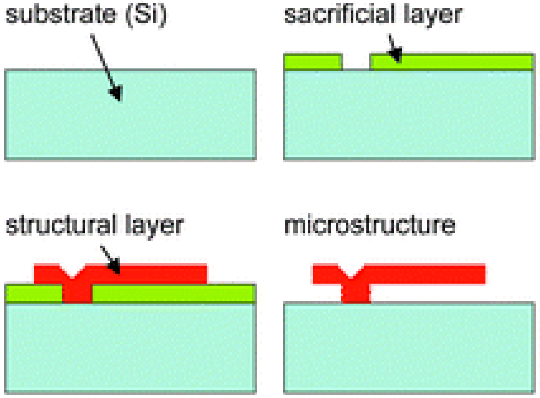


Etching the bulk silicon

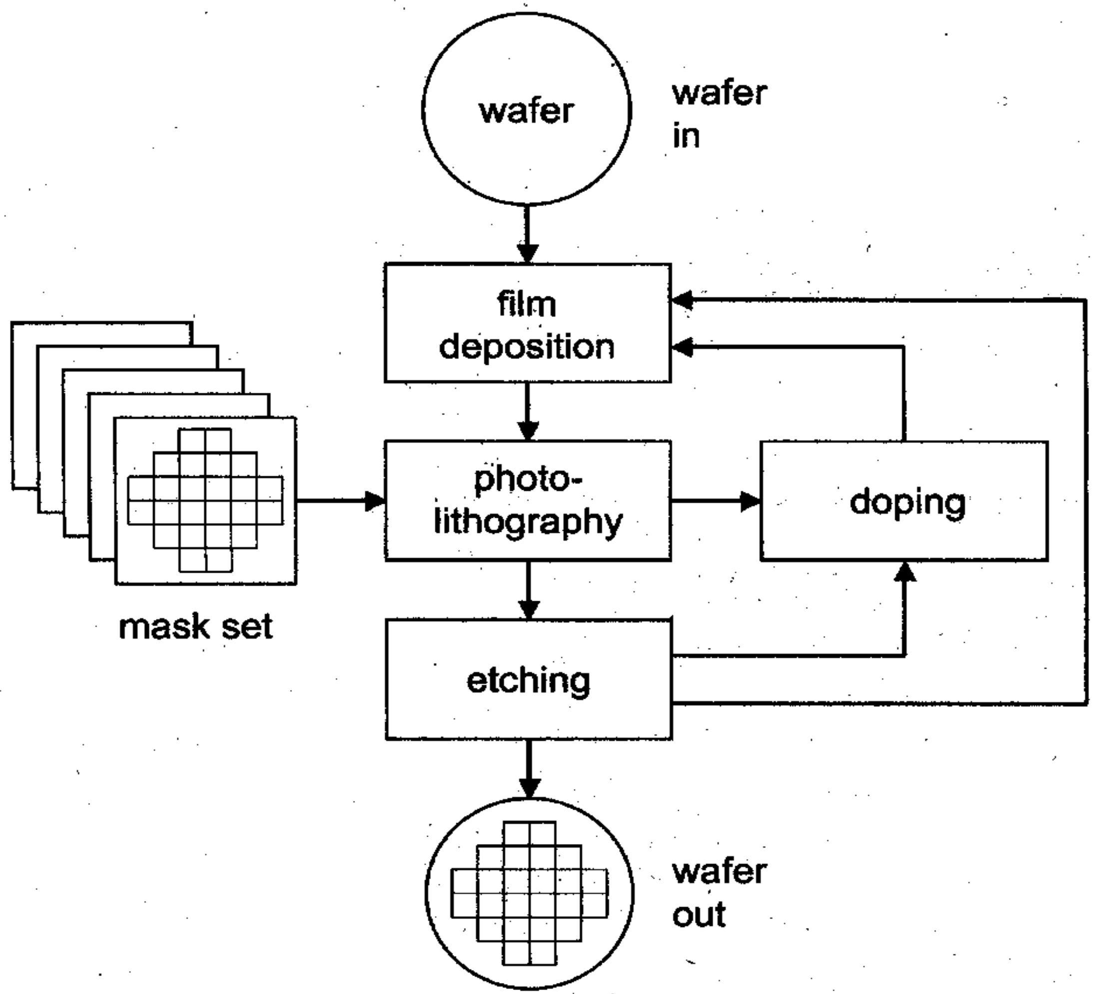


Surface Micromachining

Adding and processing layers



Microfabrication Process Flow



Basic operations:

- Cleaning
- Thin film deposition
  - from solution
  - vapor deposition
- Patterning
  - Coarse masks
  - Photolithography
- Etching
- Bonding/sealing

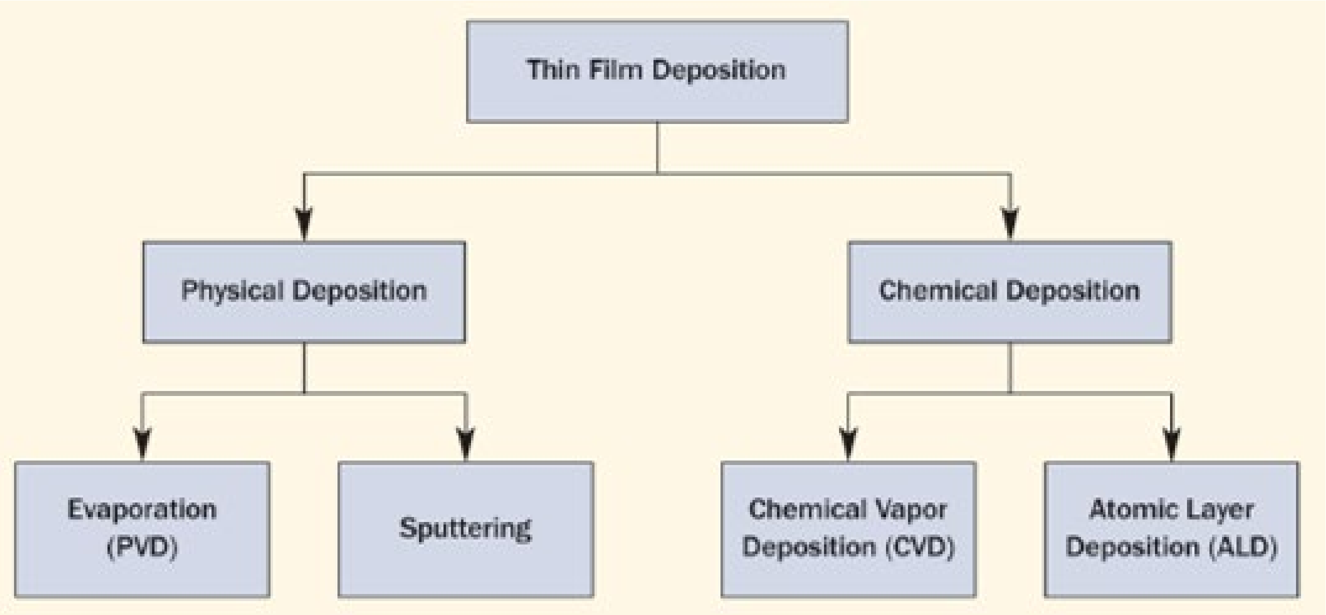
Thin Film Deposition Techniques

From solution:

- Spin coating

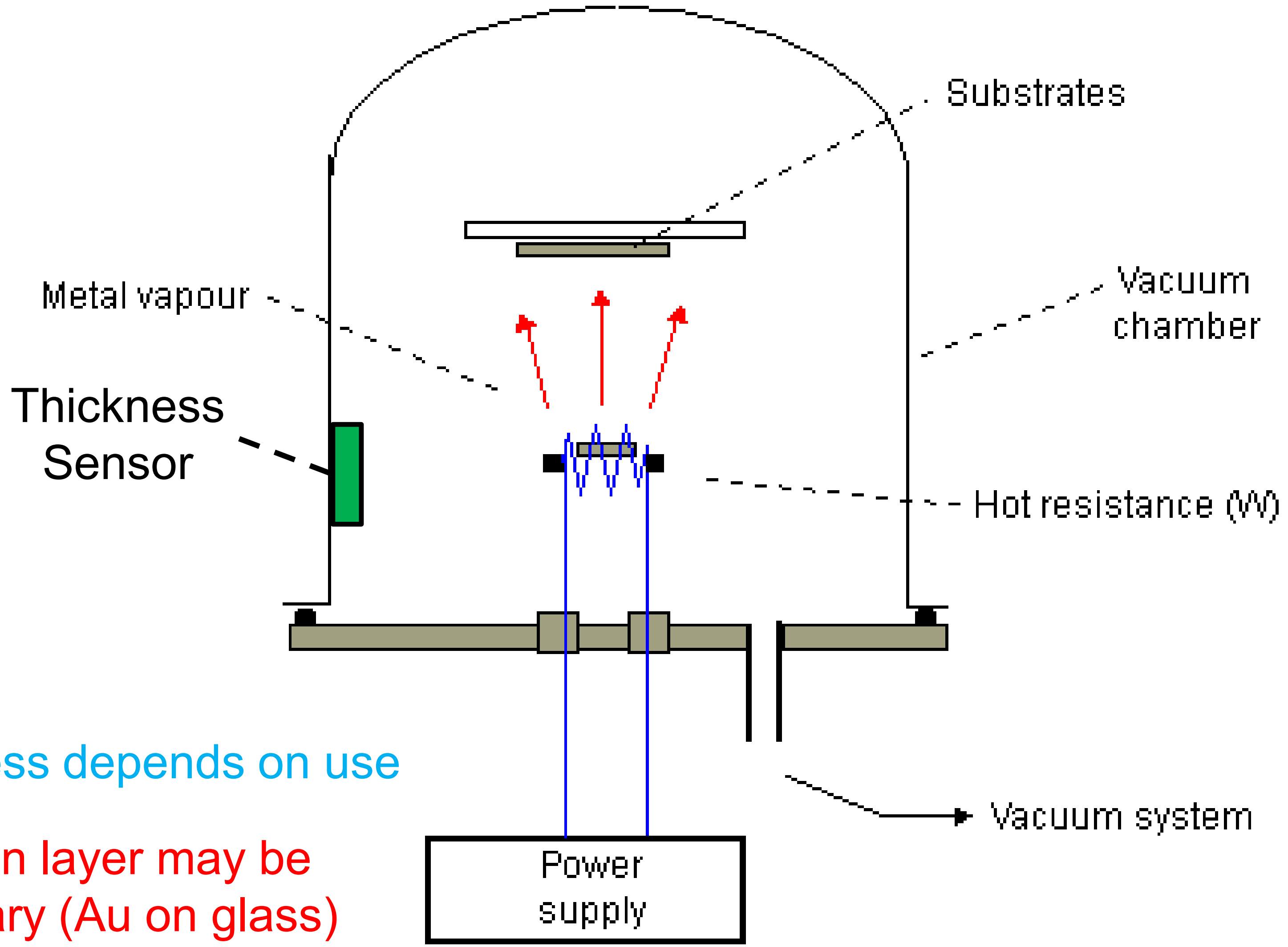
From vapor:

- Physical Vapor Deposition (PVD)
- Chemical Vapor Deposition (CVD)



PVD: Thermal Evaporation

Mainly for metals. Slow deposition rate: few Å/sec

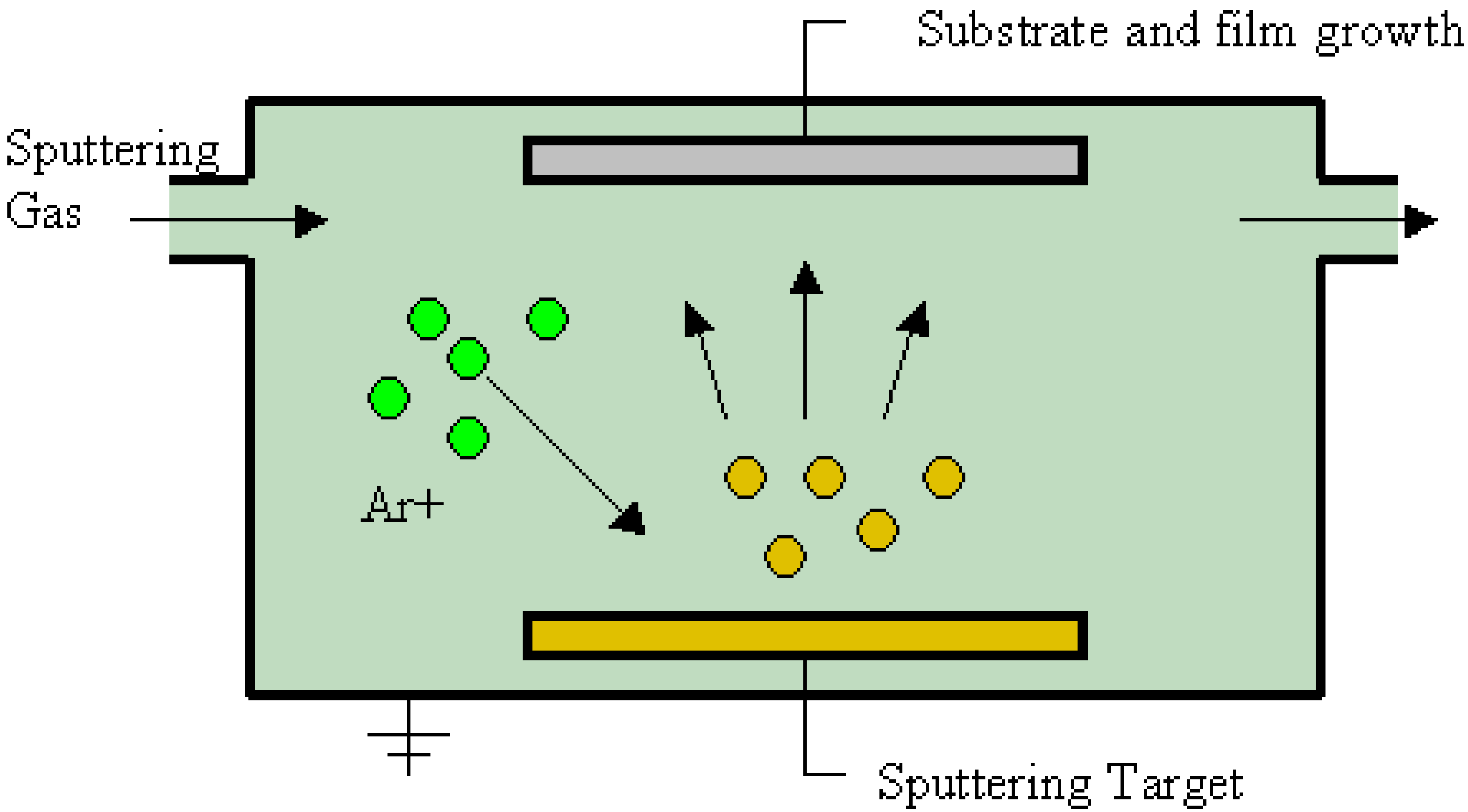


Thickness depends on use

Adhesion layer may be necessary (Au on glass)

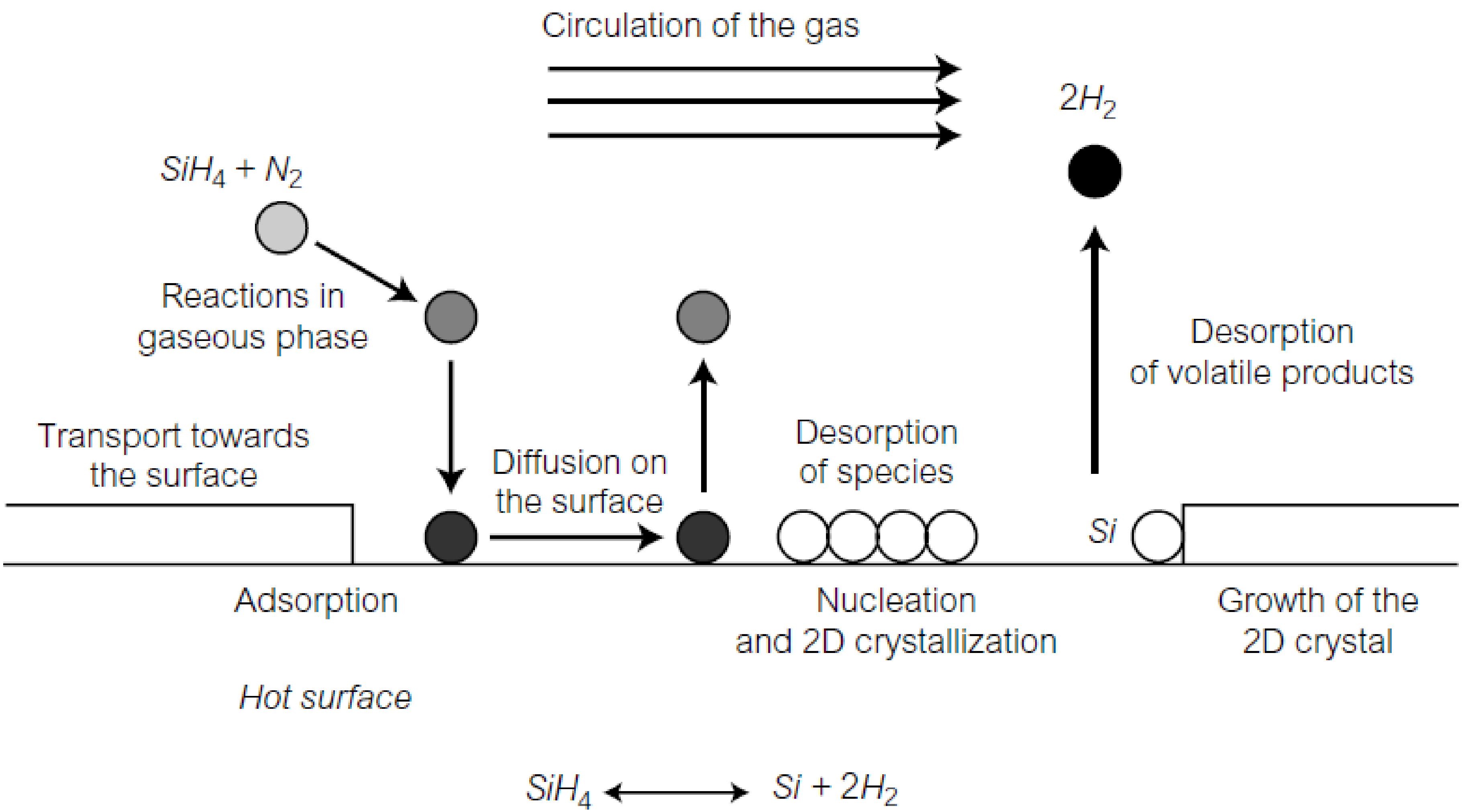
PVD: Sputtering

Use of gas ions or particles to detach atoms from the target

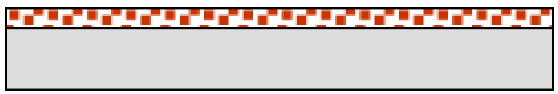


Chemical Vapor Deposition

High purity deposition of mono- & polycrystalline Si, SiO_2, Si_3N_4

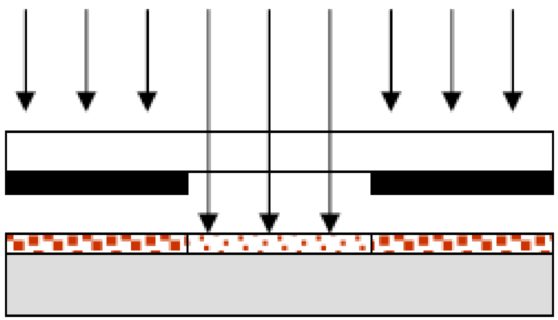


Photolithography



1. Spin-cast a photosensitive resist

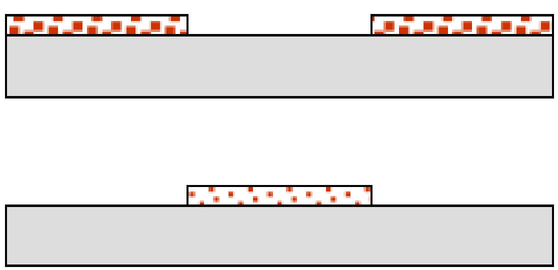
Soft bake (evaporate solvent)



2. Expose to UV light through a patterned mask aligned with a mask aligner (μm precision)

positive

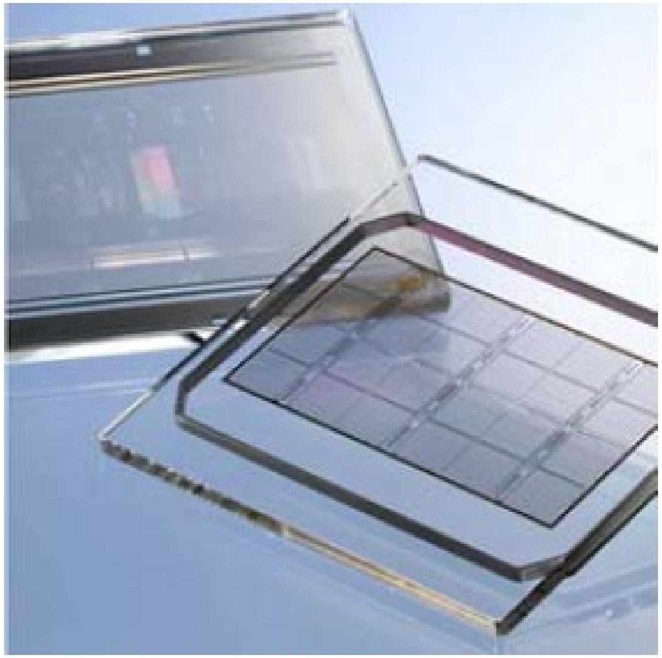
negative



3. Develop by dissolving the covered/uncovered resist regions with a proper solvent

1. Draw and Fabricate the Mask

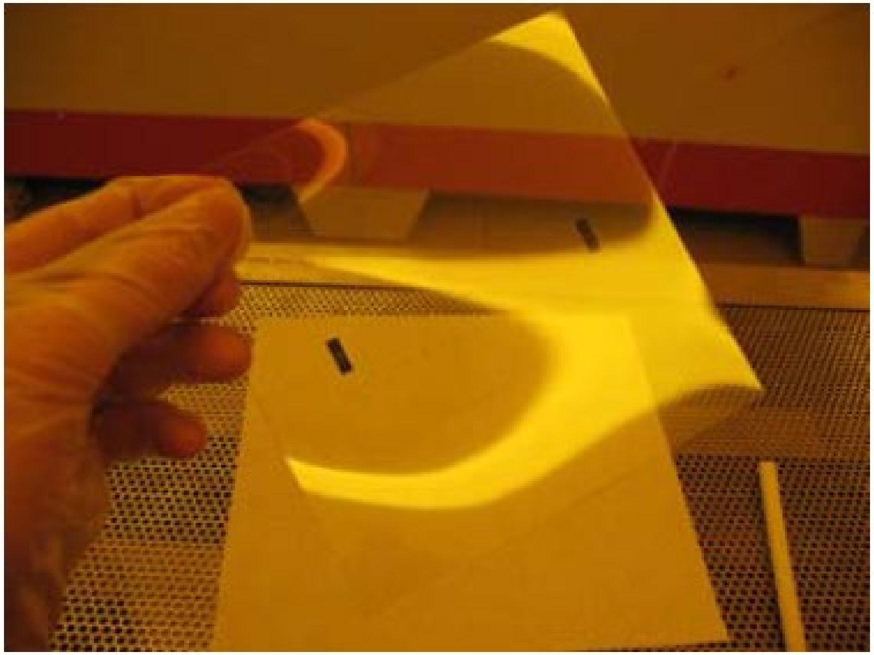
Professional



www.toppan.co.jp

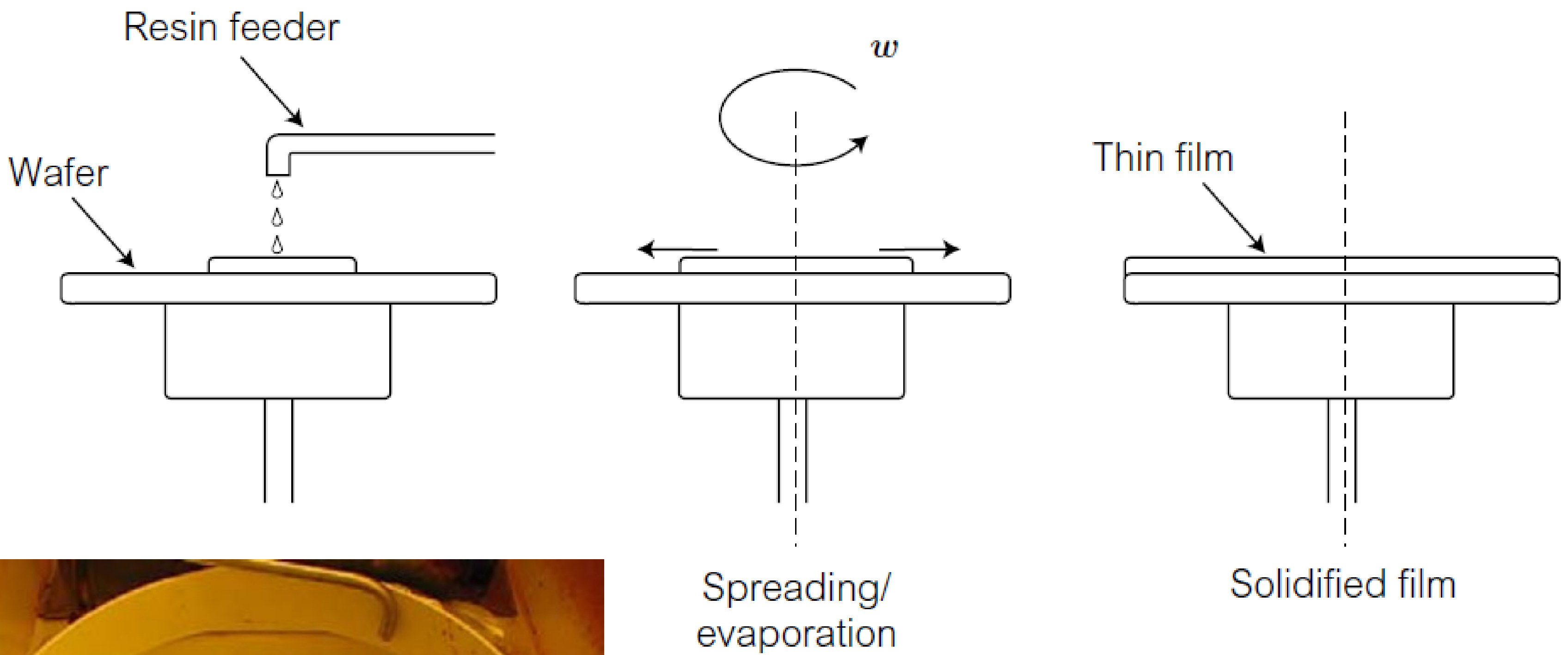
Cr patterned on quartz
with *electron beam*
(sub µm precision)

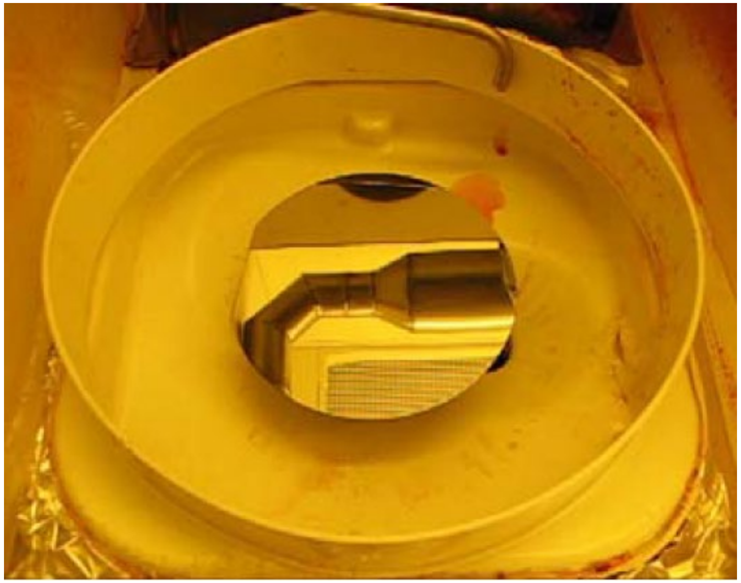
Home made



Printed on transparency film (tens of µm precision)
Fast and low-cost

2. Photoresist Deposition: Spin Coating



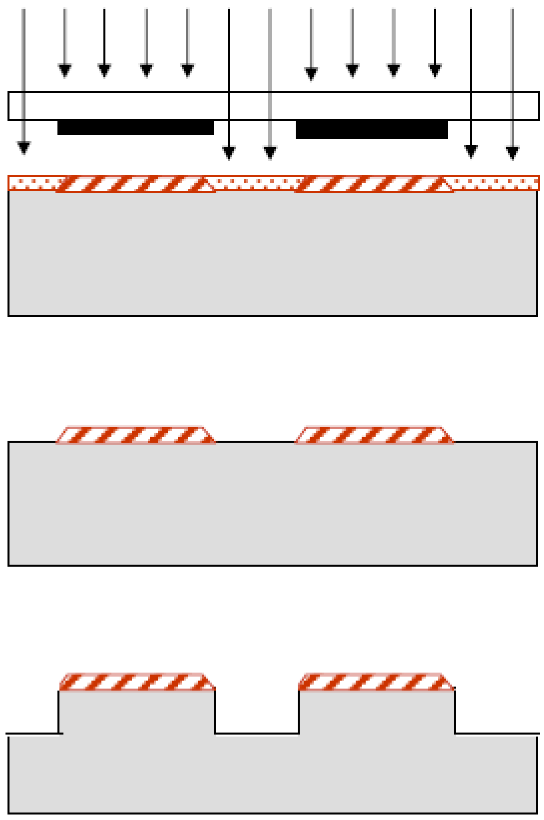


The speed ω (1000-10000 rpm) sets the final thickness (1-2µm):

$$h = kC\left(\frac{\mu}{\omega^2}\right)^{1/3}$$

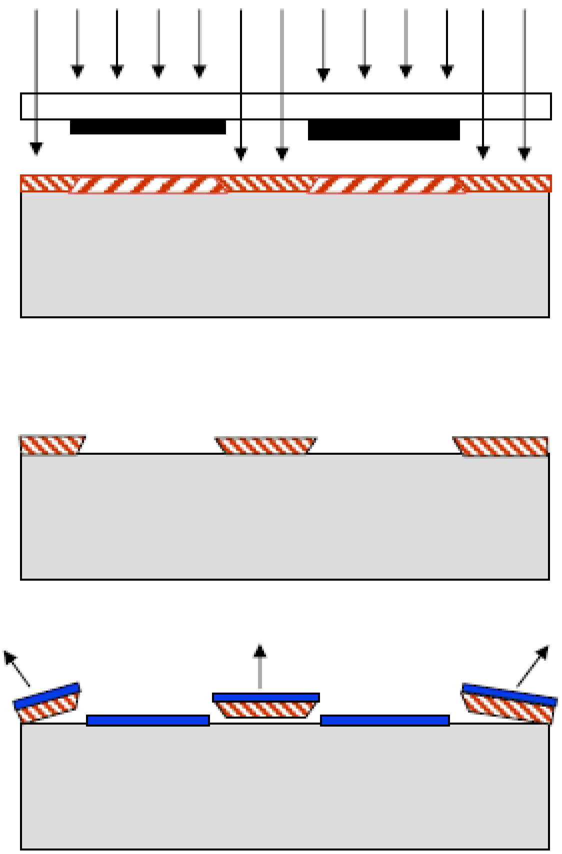
Photoresists Types

Positive



Light breaks covalent bonds

Negative



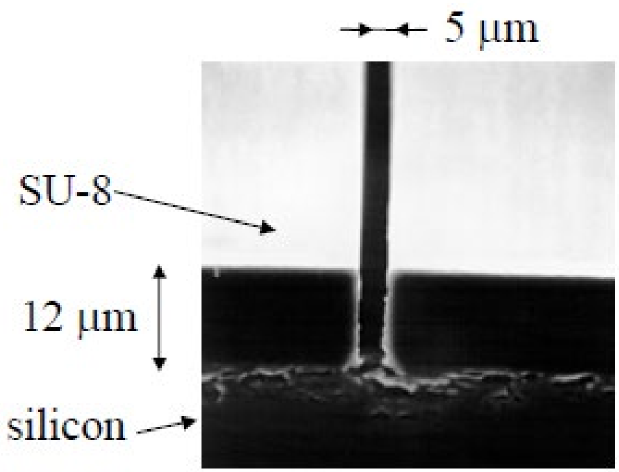
Light cross-links the polymer

Photoresist Properties

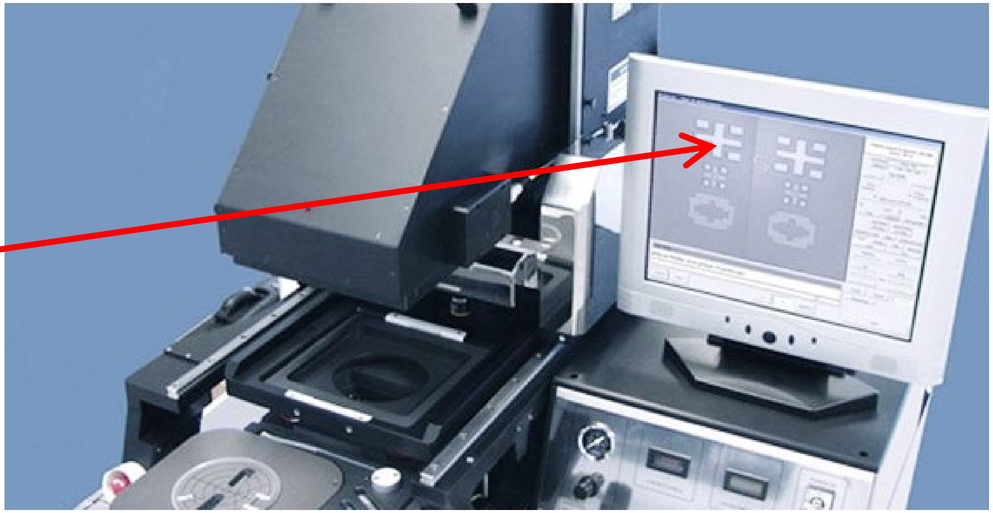
Photoresist features:

- Sensitive (homogeneously)
- Large exposed/unexposed contrast
- Wide range of thicknesses
- Selectively resistant to chemicals

Most used for biochips: **SU8** (negative) with high *aspect ratio*



3. Exposure, Etching and Hard Bake



Mask aligner:

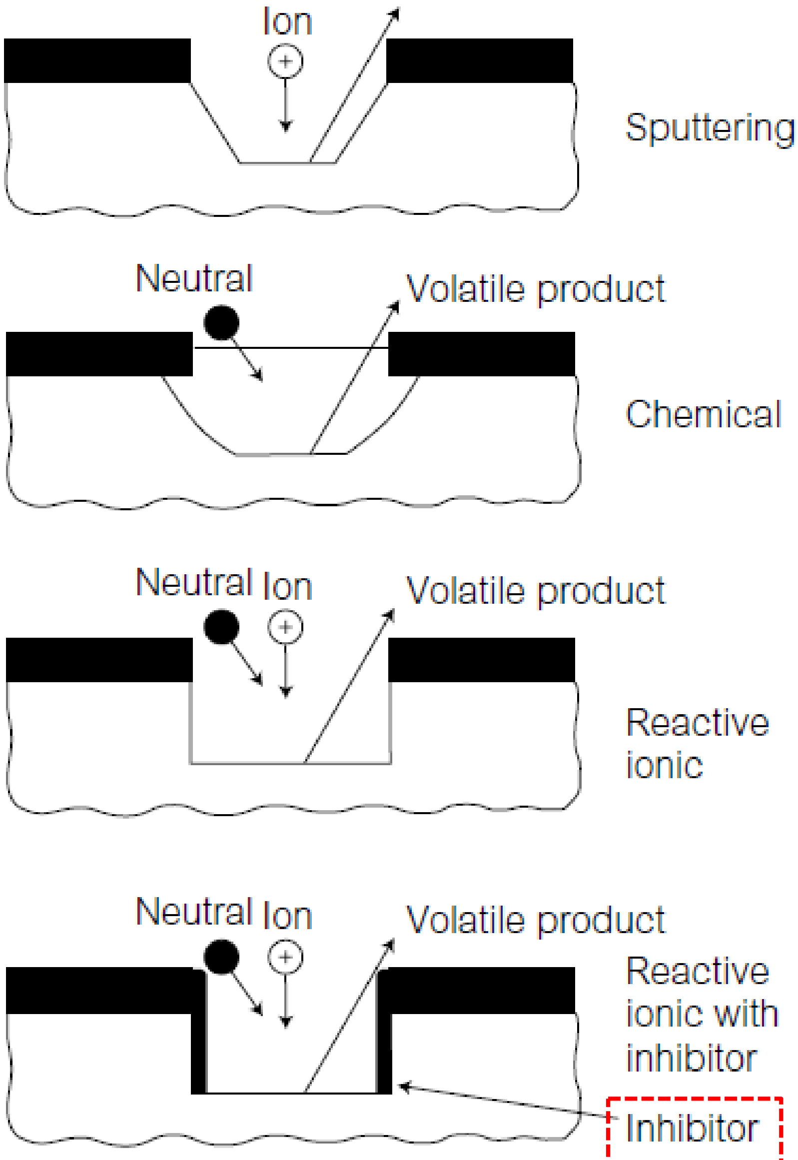
Requires alignment marks

Contact photolithography:

- Mask and substrate get degraded by contact
- Resolution limited (2-3μm) by diffraction and by the resist thickness (s): $\delta \approx 3\sqrt{\lambda s}$;

Projection photolithography: lens system to reach sub-μm

Dry Etching



Physical removal of material

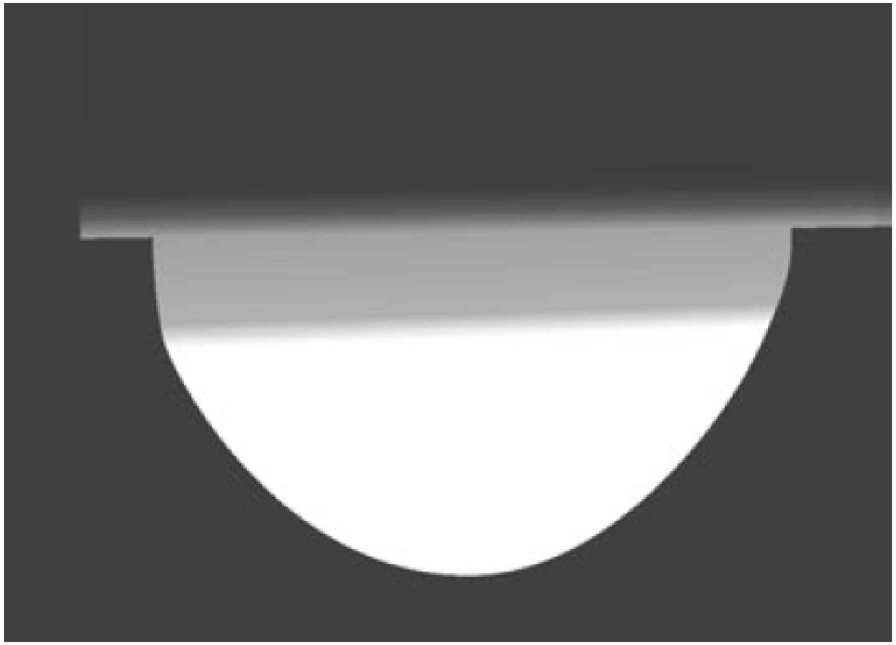
Chemical removal of material

Physical and chemical

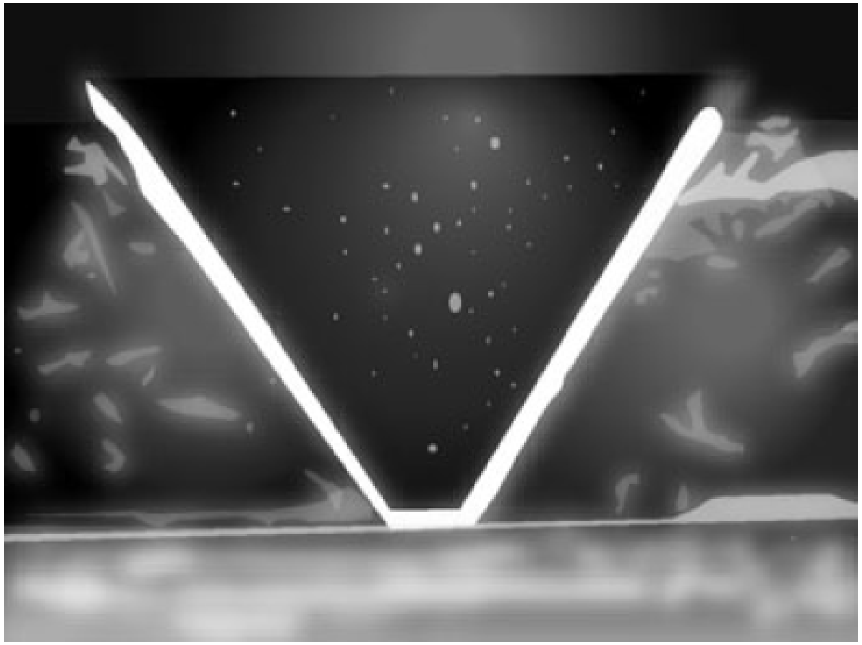
High aspect ratio

Wet Etching

Isotropic



Anisotropic

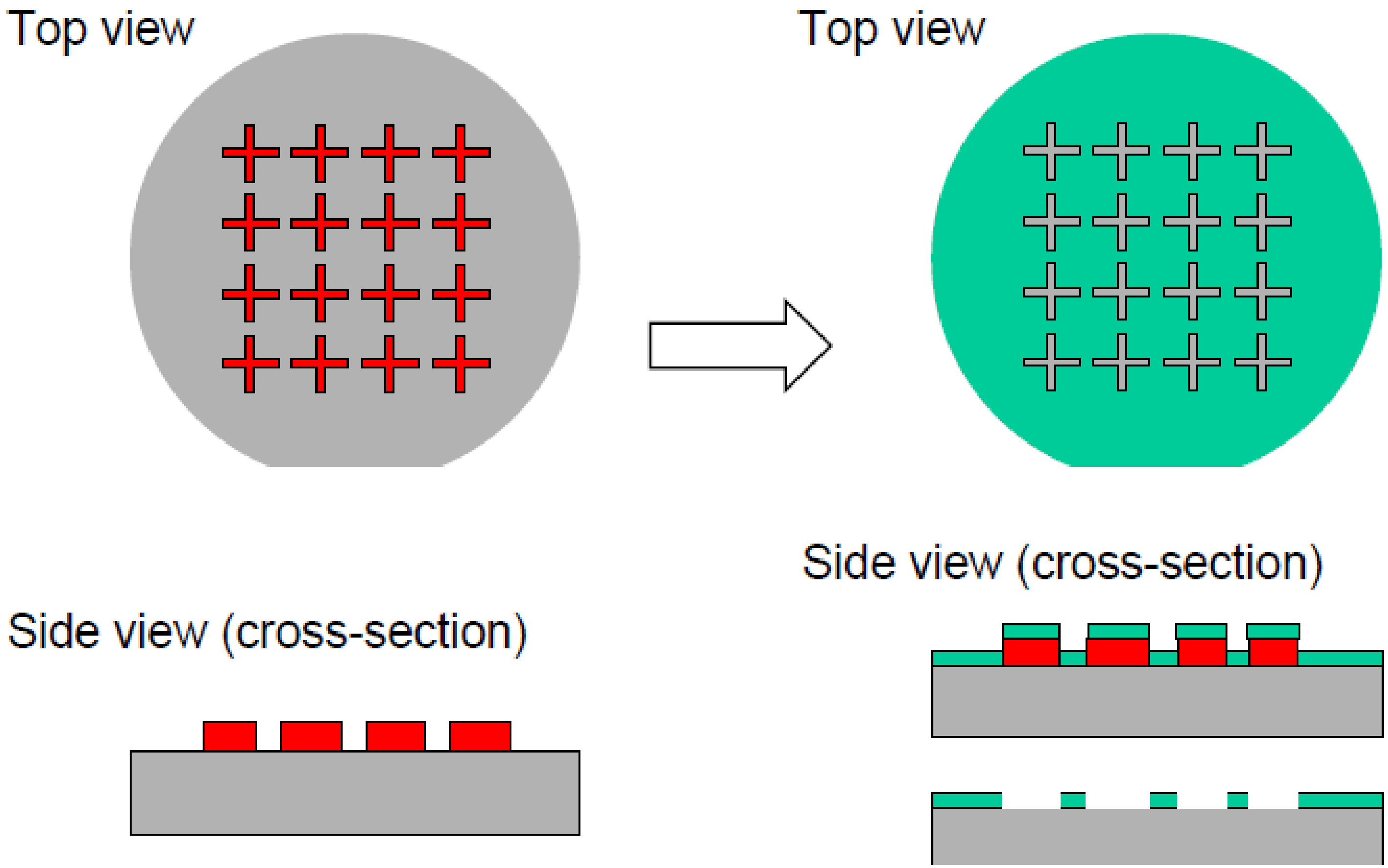


Different each rates for crystallographic planes

Rates: µm/hr to µm/sec

Lift-off (Mainly for Metals)

1. Pattern sacrifical resist layer
2. Deposit uniformly the metal
3. Etch the resist



On-Line Learning Resources

Several videos of microfabrication are available on-line:

Intro to microfabrication photolithography
https://www.youtube.com/watch?v=B0g1_rr-ApI

Spin Coating: https://www.youtube.com/watch?v=uv1A7qAqgRk

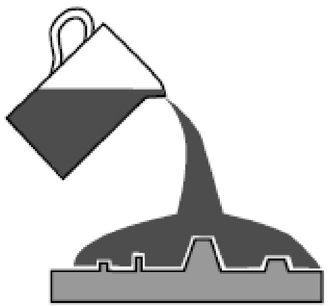
Evaporation: https://www.youtube.com/watch?v=f7UxBawRPj4

Sputtering: https://www.youtube.com/watch?v=GMbj8FUDEQ0

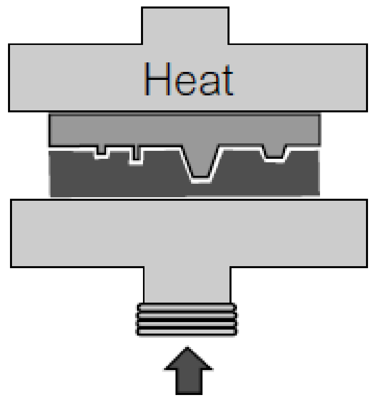
Silicon vs Plastic

	Silicon	Glass/Fussed Silica	Plastics
Thermal conductivity (cal/cm×s×°C)	0.35	$\sim 2 \times 10^{-3}$	4.5×10^{-4}
Bioassay compatibility	fair (oxide/nitride surface layer)	fair	very good
Optical detection	visible/UV: strong absorbance IR: transparent	glass: very good fused silica: excellent	poor to very good (varies according to polymer choice and wavelength)
Microfabrication	many well-developed approaches	isotropic wet etching only	silicon or glass mastering plus replication techniques; direct methods (ablation, dry etching)
Feature aspect ratio (depth: width of microchannel)	<0.1 - 40	<0.5	dependent on master for replication methods
Manufacturing methods	well developed	need development	well developed
Cost	inexpensive (small single devices) to expensive (large-area device arrays)	moderately expensive	inexpensive

Plastic Fabrication

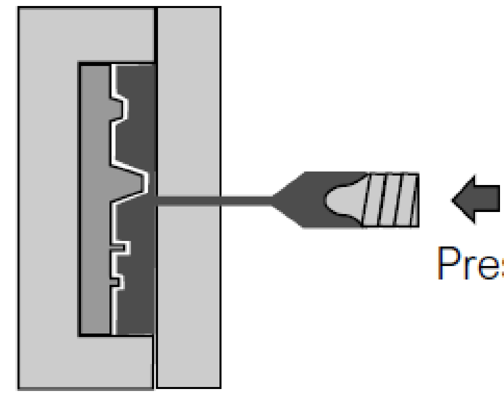


Molding



Casting

Hot Embossing



Injection molding

Manual

PDMS

Solid plastic softened with temperature

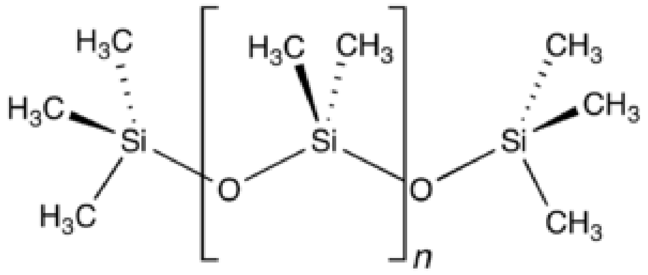
Liquid plastic (PMMA)

Industrial

Most Widely Used Material: PDMS

Polydimethylsiloxane (C, Si, O)

Hydrophobic elastomer

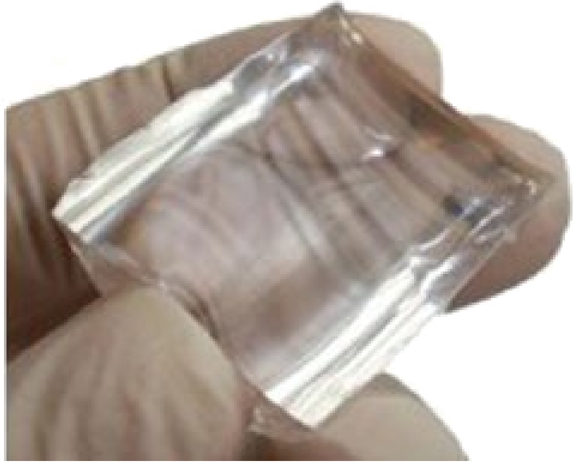


Advantages:

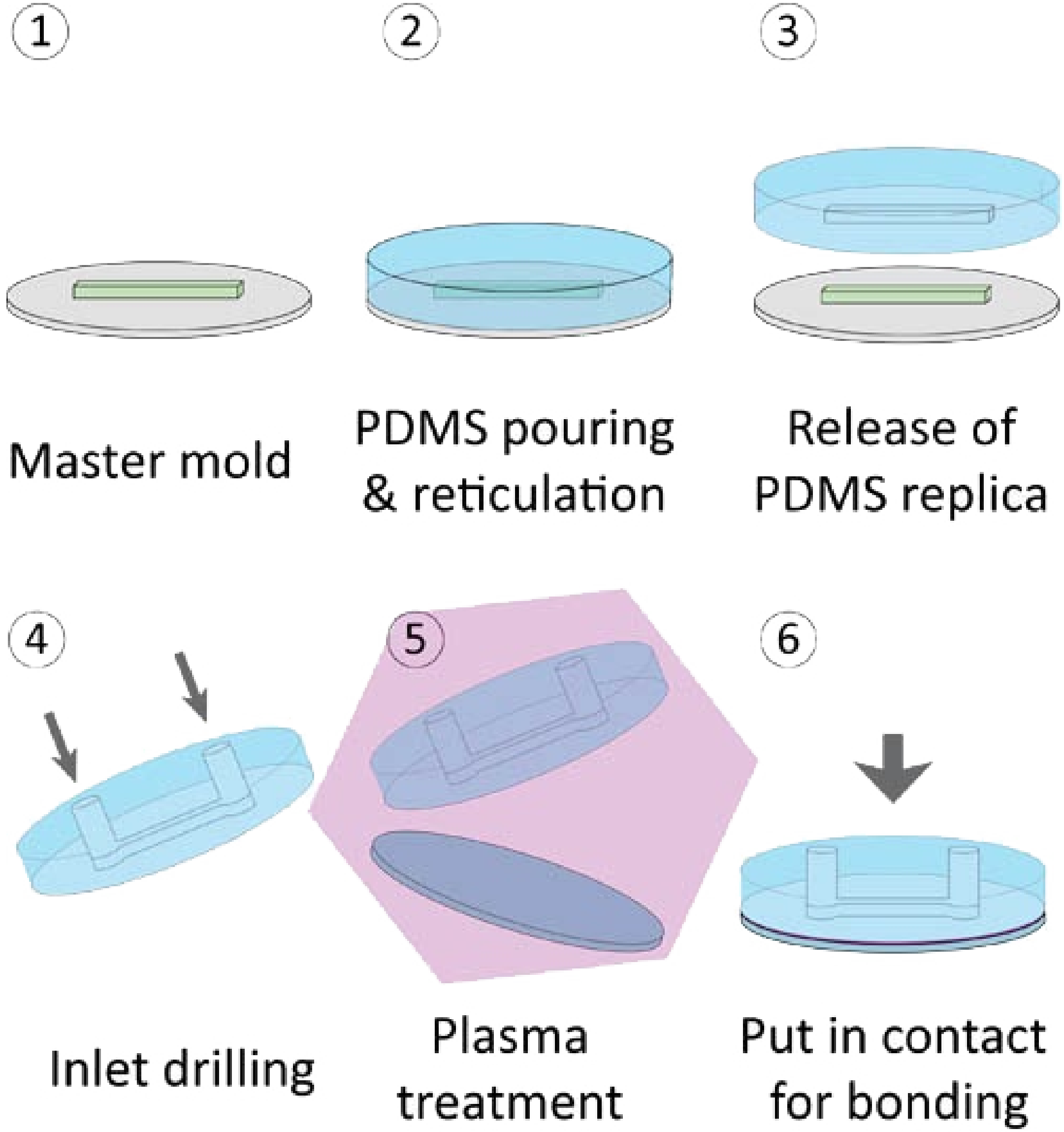
- Transparent at optical frequencies (240nm-1100nm),
- Biocompatible
- Permeable to gases
- Pourable from liquid phase (casting and spin coating)

Issues:

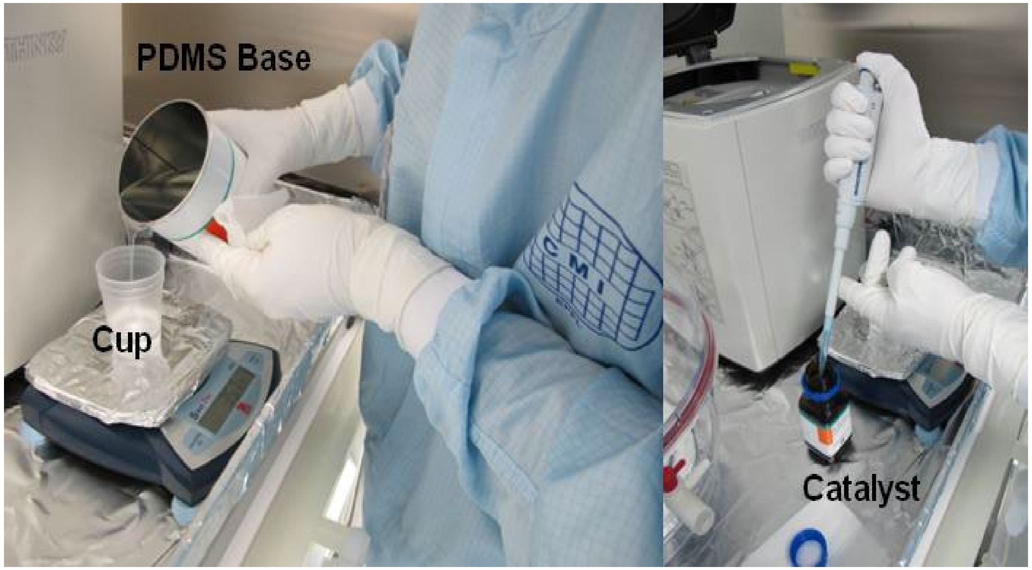
- Aging
- Bad substrate for metal deposition
- Chemical resistance

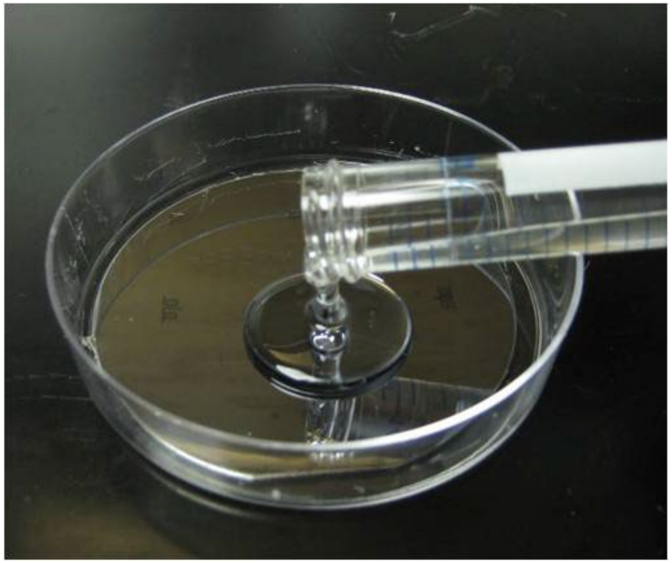


PDMS Molding Sequence



In Practice Steps: Mixing

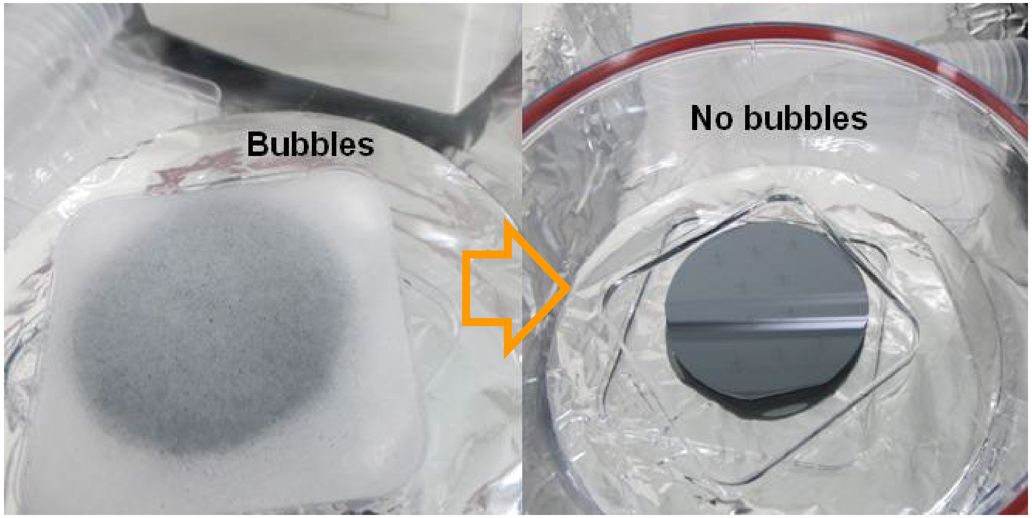


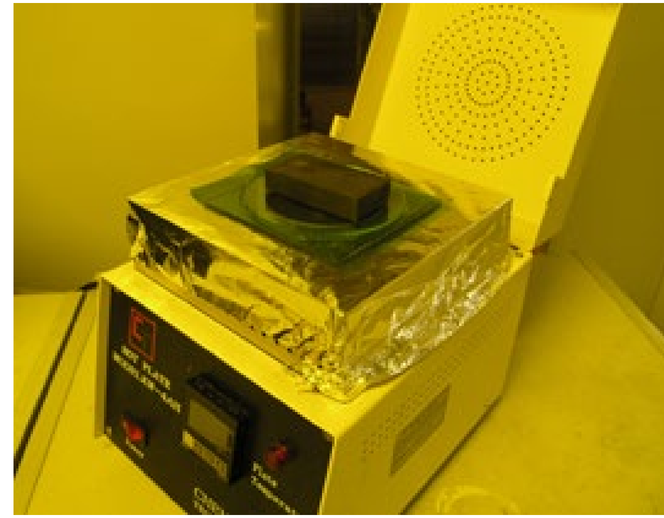


1. Mix elastomer base and curing agent (10:1)

2. Pour into the mold

In Practice Steps: Curing

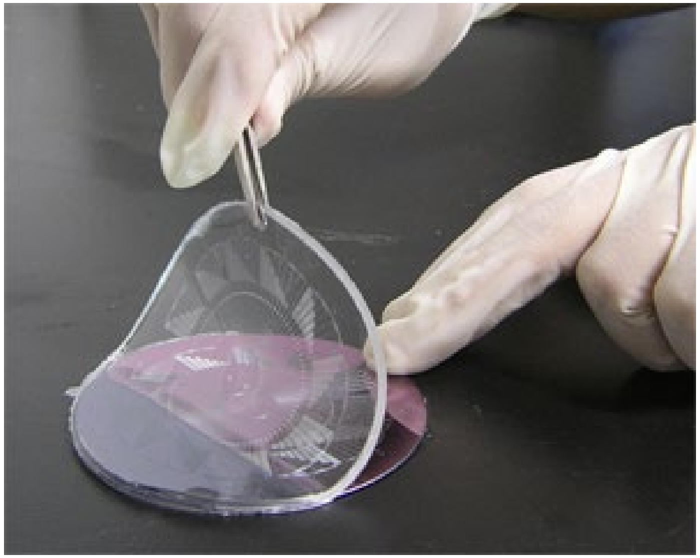




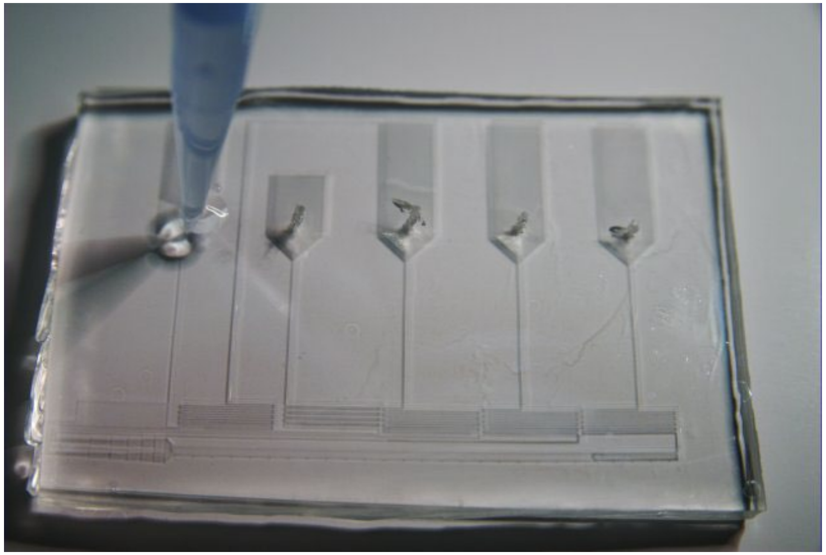
3. Degas to remove bubbles

4. Cure at 80°C

In Practice Steps: Detaching



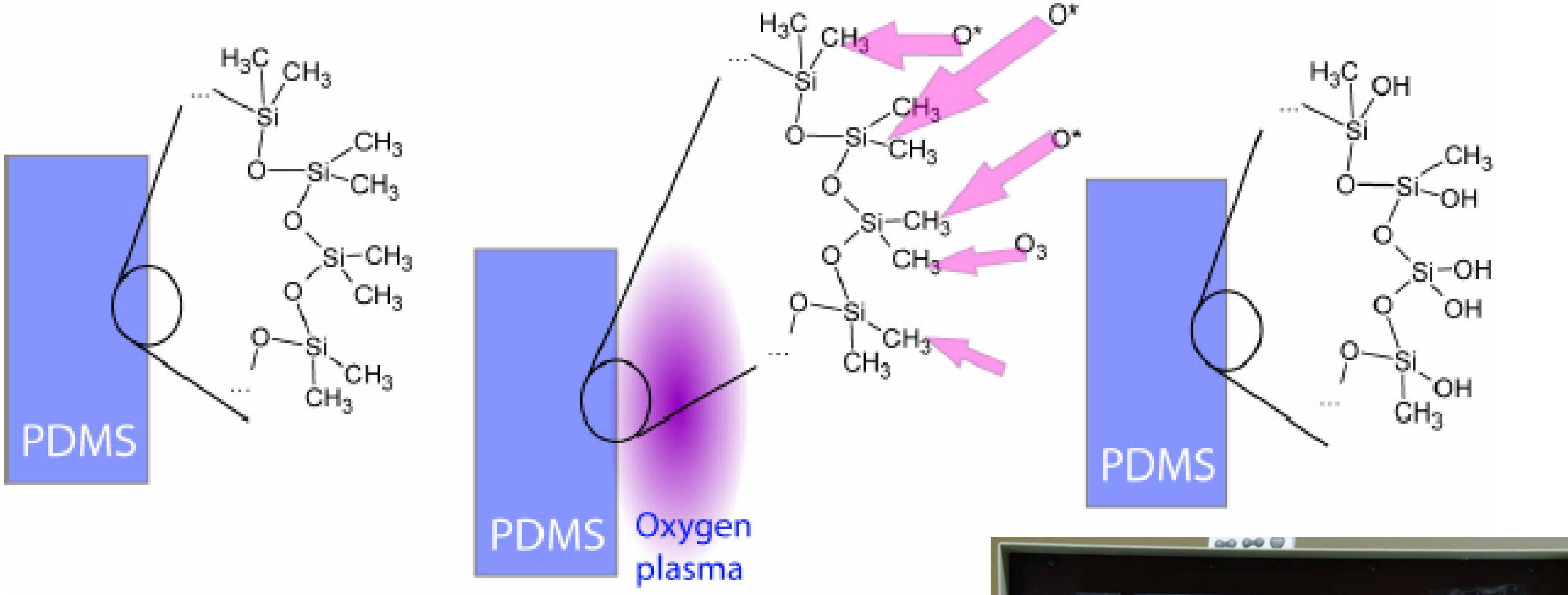
5. Peel off



6. Puch-through holes

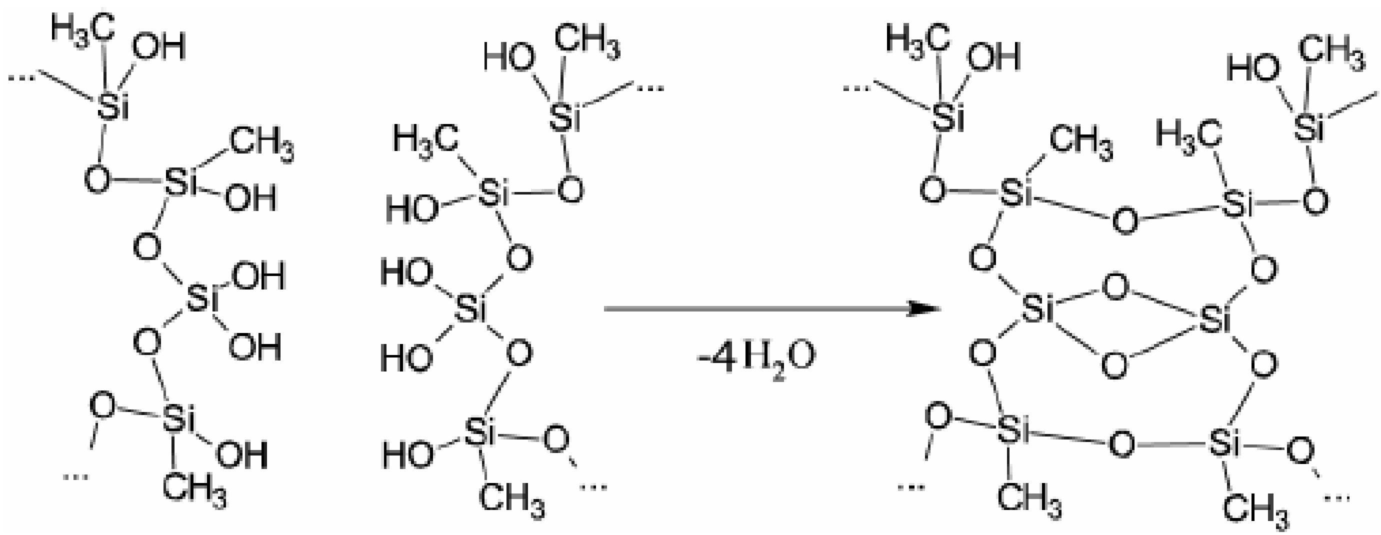
In Practice Steps: Surface Activation

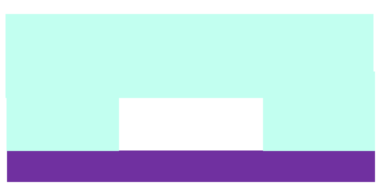
7. Use oxigen plasma to activate the surfaces (break some bonds) for permanent bonding between glass and PDMS



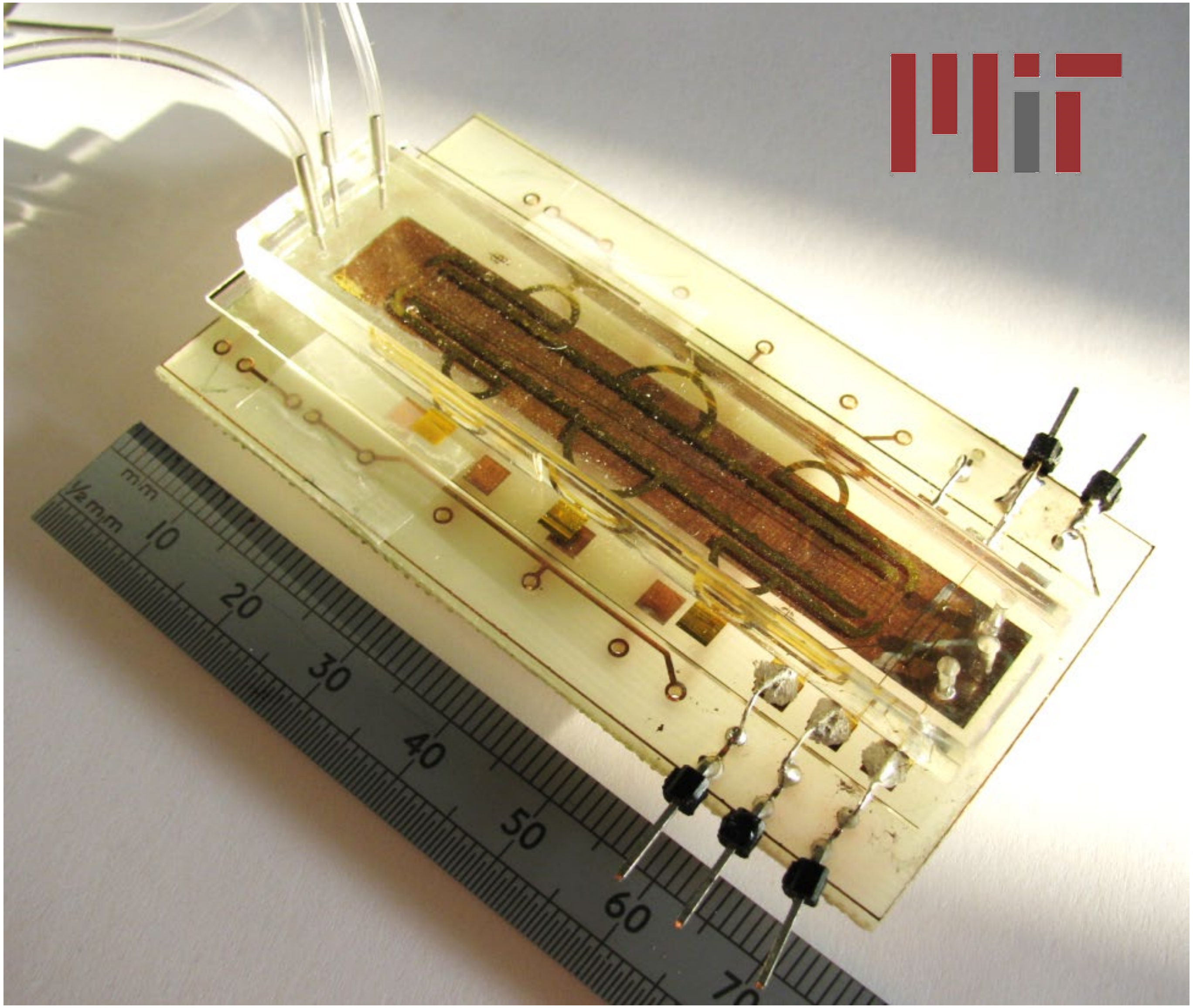
http://www.ims.ut.ee/~alar/microtech/Ch1_5/

8. Bond the glass substrate with the PDMS slab: **covalent bond**





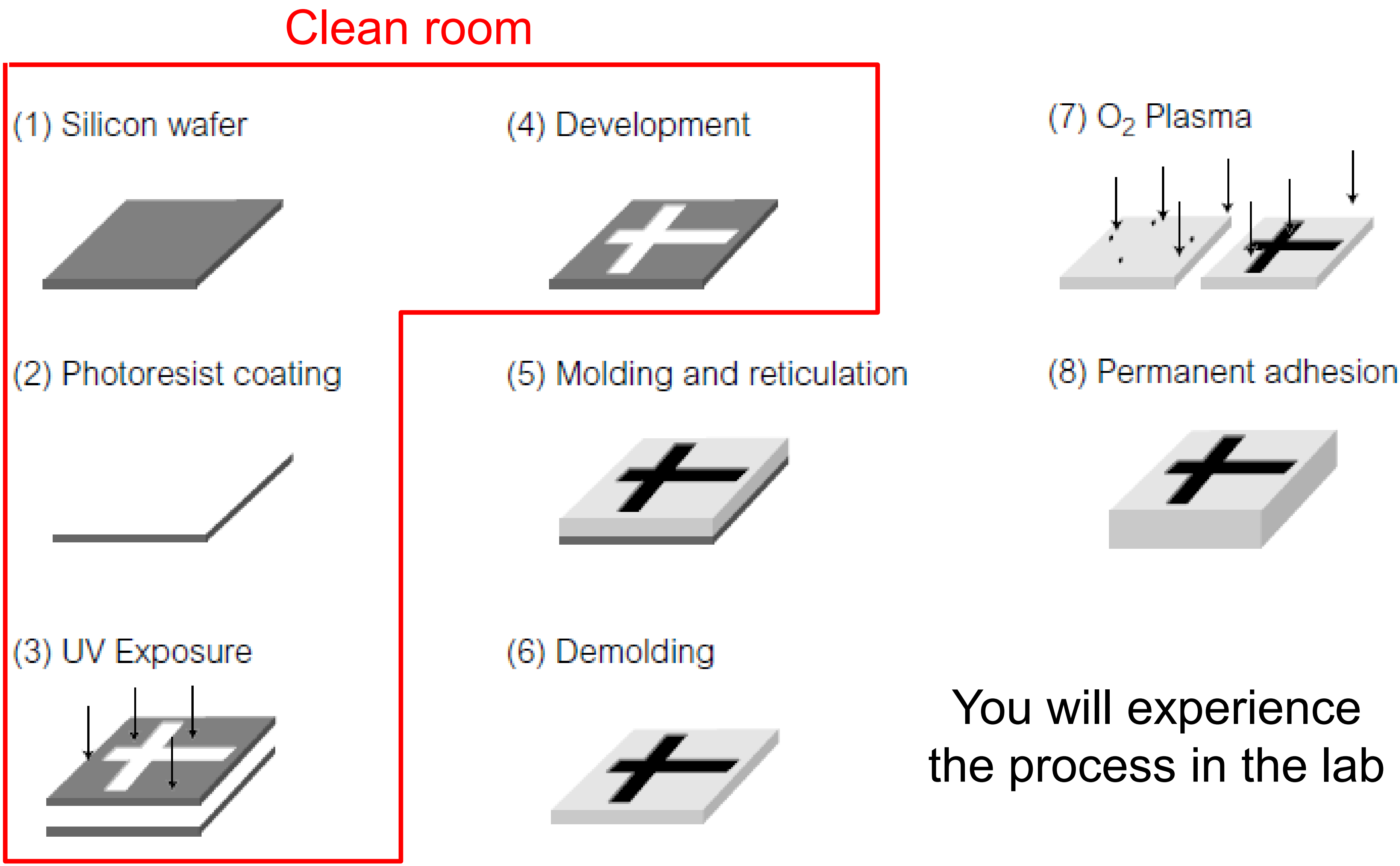
Final Result: Assembled Device



8. Electrical connections

9. Fluidic connections

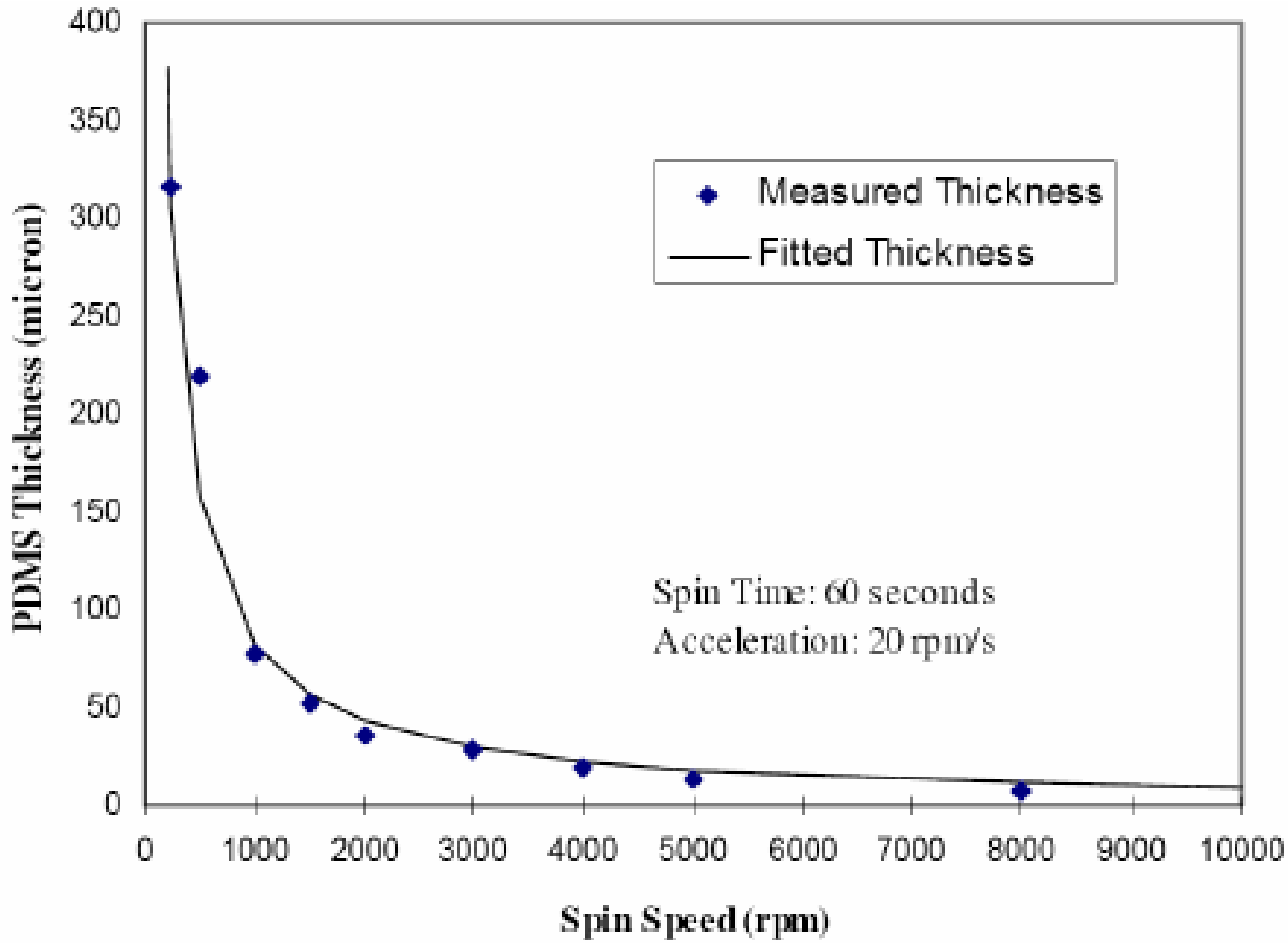
The Full Process



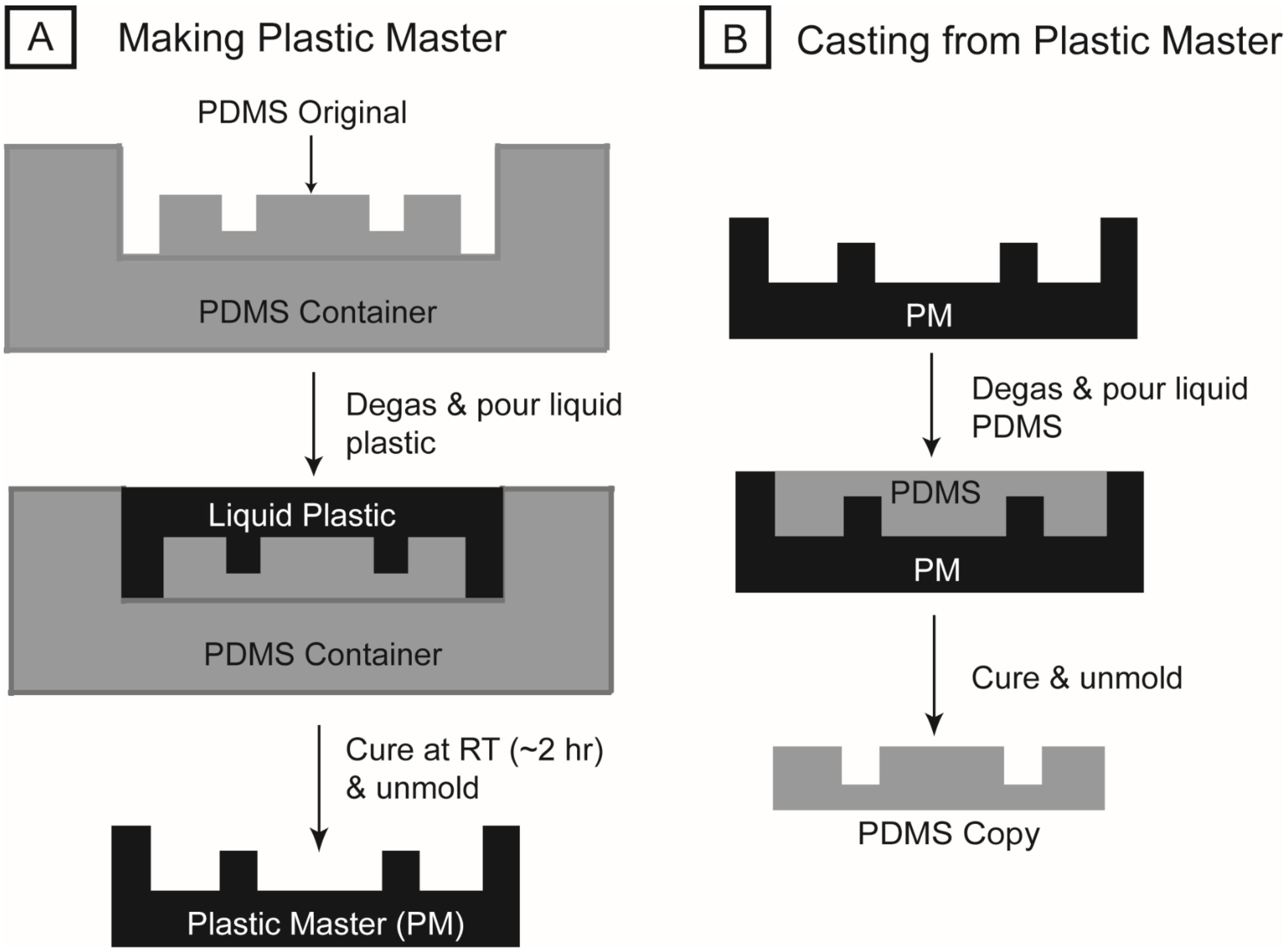
You will experience the process in the lab

PDMS can be Spun

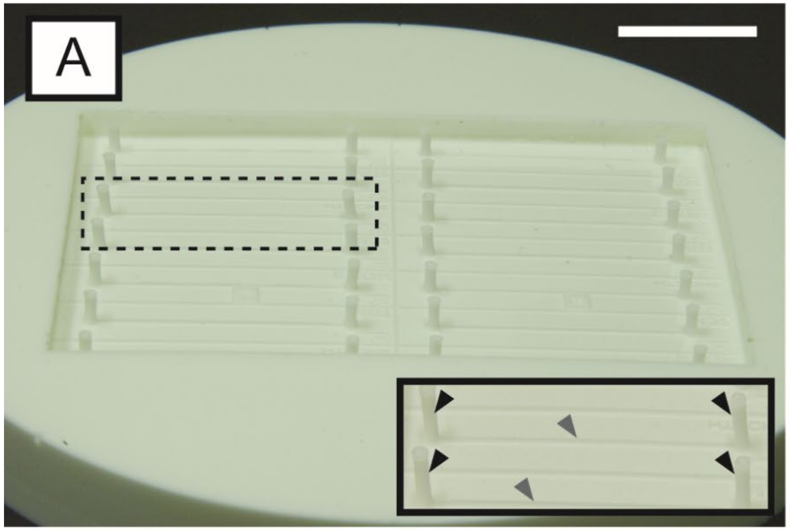
In order to obtain **thin layers**, PDMS can be spun:

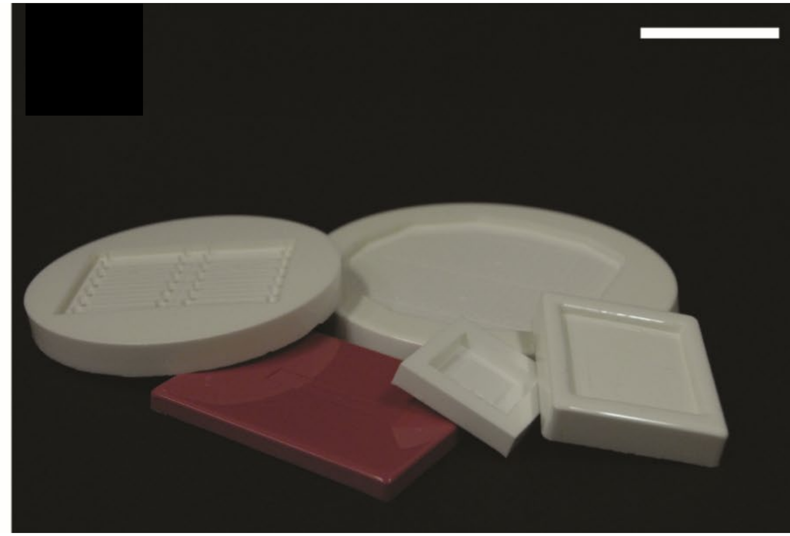


Plastic Molds to Preserve Silicon Molds



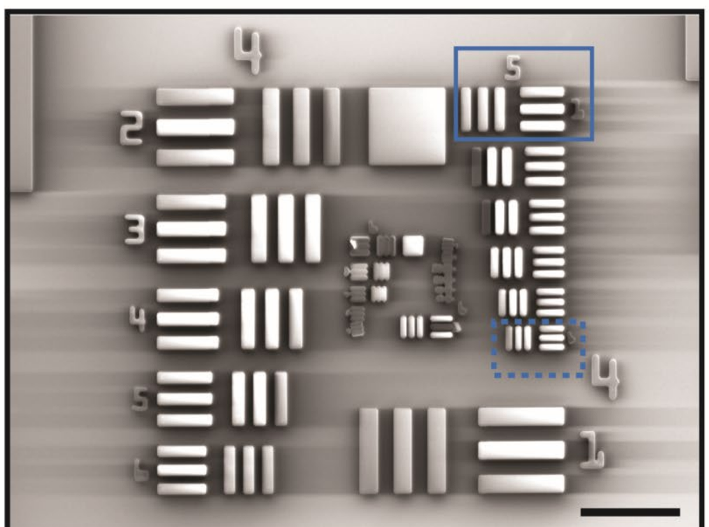
Examples

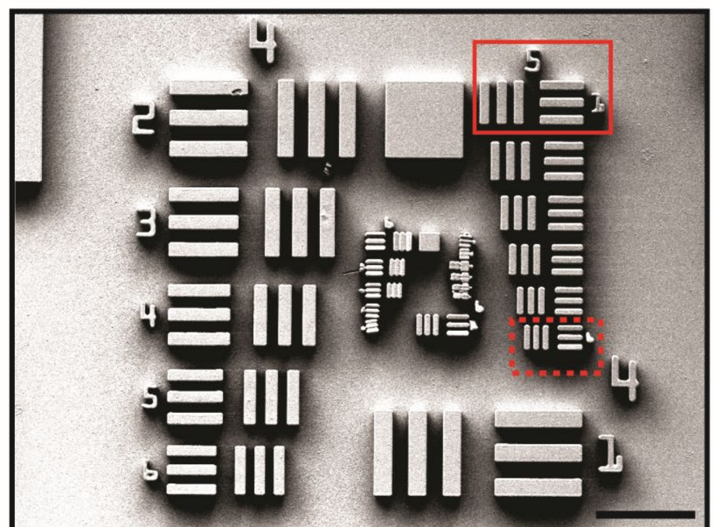




A Silicon and SU-8 Master (SPM)

B Plastic Master (PM)





Other Plastics: Soft vs. Rigid

	PDMS	PC	PETE	PI	PMMA
Density ρ (kg/m^3)	1300	1200	1370 - 1380	1430	1170 - 1200
Tensile strength (MN/m^2)	-	52 - 62	66	68	45 - 72
10^5 x Coeff. linear expansion (°C)	8 - 30	6.6	6	4 - 5	5 - 9
Heat Capacity (J/g×K)	-	1.17 - 1.25	1.25	1.12	1.5
Thermal Conductivity (W/m×K)	0.15 - 0.32	0.19	0.29	-	0.17 - 0.25
Refractive index	1.43	1.59	1.54	translucent	1.48 - 1.50
Resistivity (Ω×cm)	10^{14} - 10^{15}	10^{16}	10^{16}	>10^{16}	>10^{14}
Dielectric constant	2.7	2.97 - 3.17	3.25	3.4	3.3 - 4.5
	Elastic				Rigid

Alternatives to PDMS

Other polymers have been identified as complementary to PDMS, with similar fabrication procedures, but with higher rigidity (high pressure) and better resistance to solvents:

- Thermoset Polyester (TPE),
- Polyurethane Methacrylate (PUMA)
- Norland Adhesive 81 (NOA81)

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TUTORIAL REVIEW

Rapid prototyping polymers for microfluidic devices and high pressure injections†

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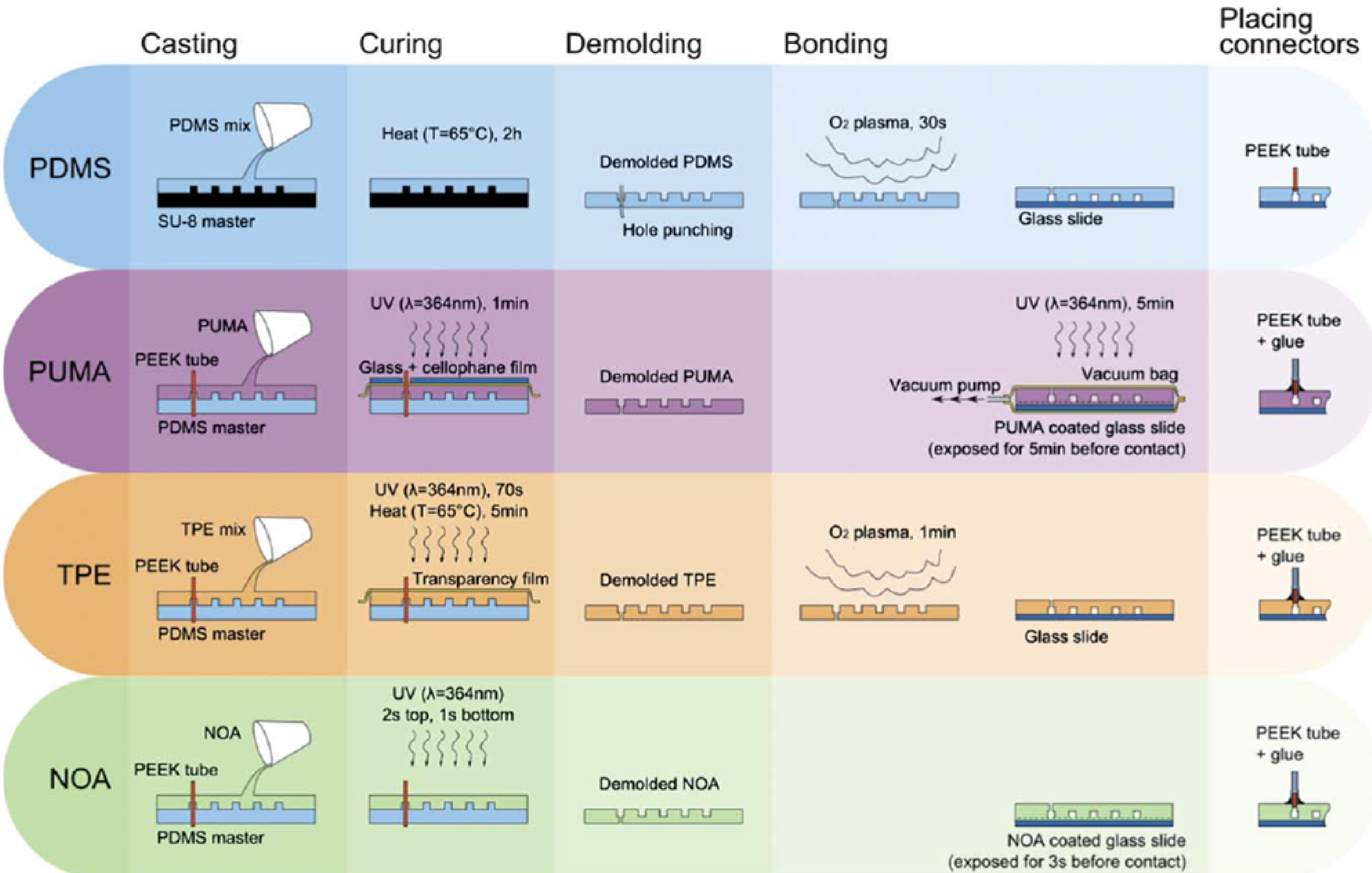
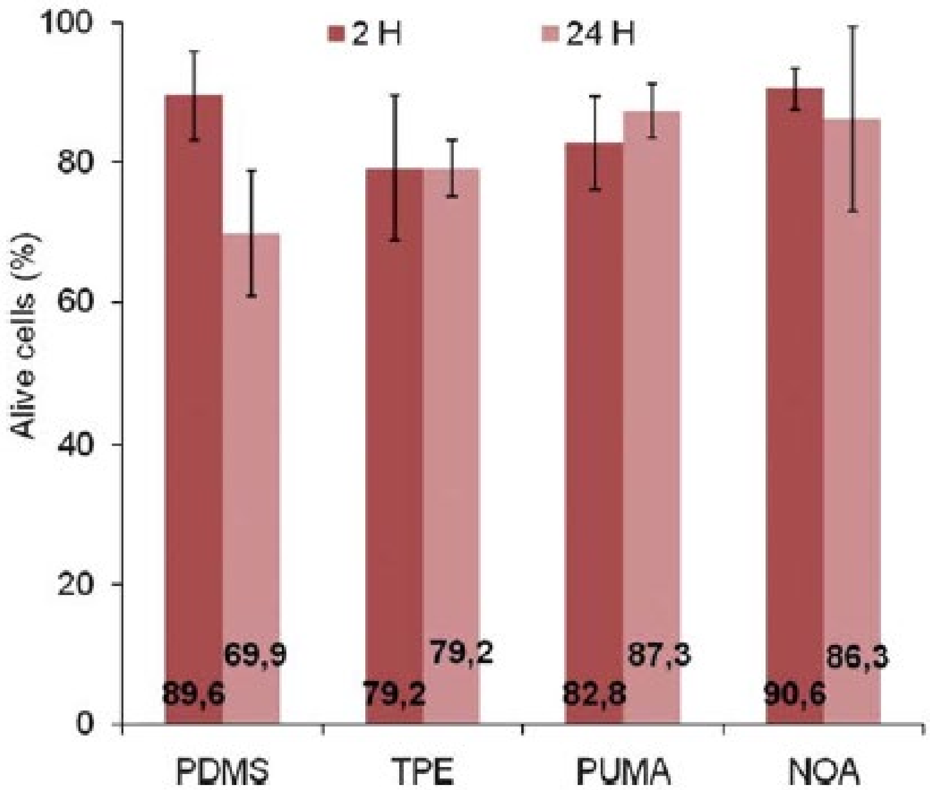
Alternatives to PDMS



Biocompatibility

For cell-hosting biochips, complex to define and assess:

1. Cell viability is preserved
2. No genetic/metabolic modification is observed



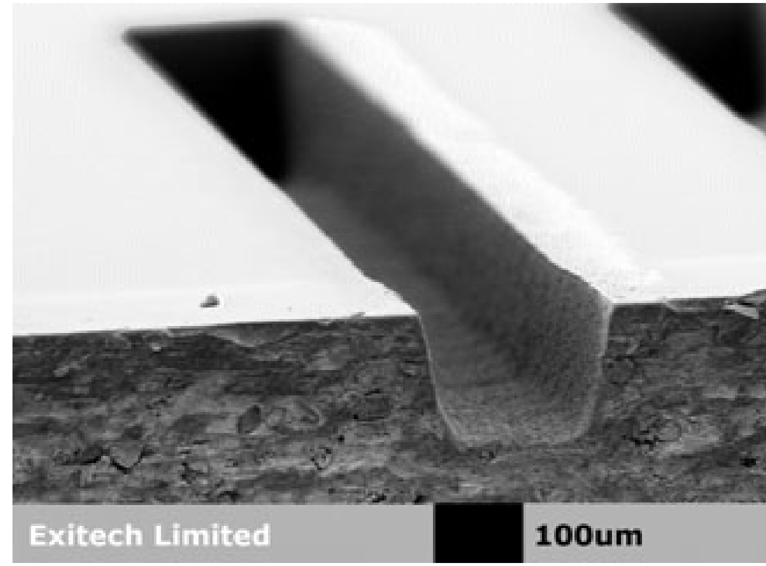
Comparison Conclusions

Table 2 Overall performance of PDMS, TPE, PUMA and NOA as materials for rapid prototyping of microfluidic devices and high pressure injections

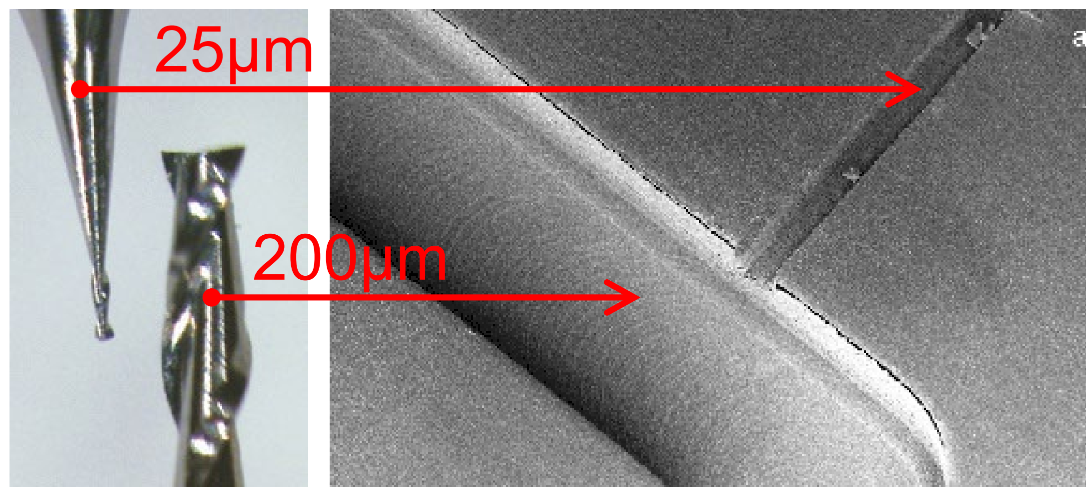
		PDMS	TPE	PUMA	NOA
Material properties	Hardness	–	++	+	+
	Biocompatibility	+	–	++	++
	Optical Transparency	++	+	+	+
	Solvent Compatibility	–	+	+	+
Performance for Fast Prototyping and High Pressure Injections	Ease of Use	+	–	++	++
	Fabrication Time	1h + 2h	1h + 1day	1h + 2h	1h + 2h
	Cost	<$1/mL	~$4.5/mL	~$2.2/mL	<$1/mL
	Replication Fidelity	+	+	+	+
	Channel Deformation	–	++	+	+
	Stabilization time	–	++	++	++
	Maximum Pressure	–	++	+	+
	Inertial Focusing	–	++	+	+

Direct Micromachining of Plastic

- **Laser ablation:** UV and femtosecond power laser
  plastic sublimation
  ~µm resolution
  200-500nm roughness



- **Micromilling:**
  hundreds of µm channel
  a few µm resolution



http://www.minitech.com/

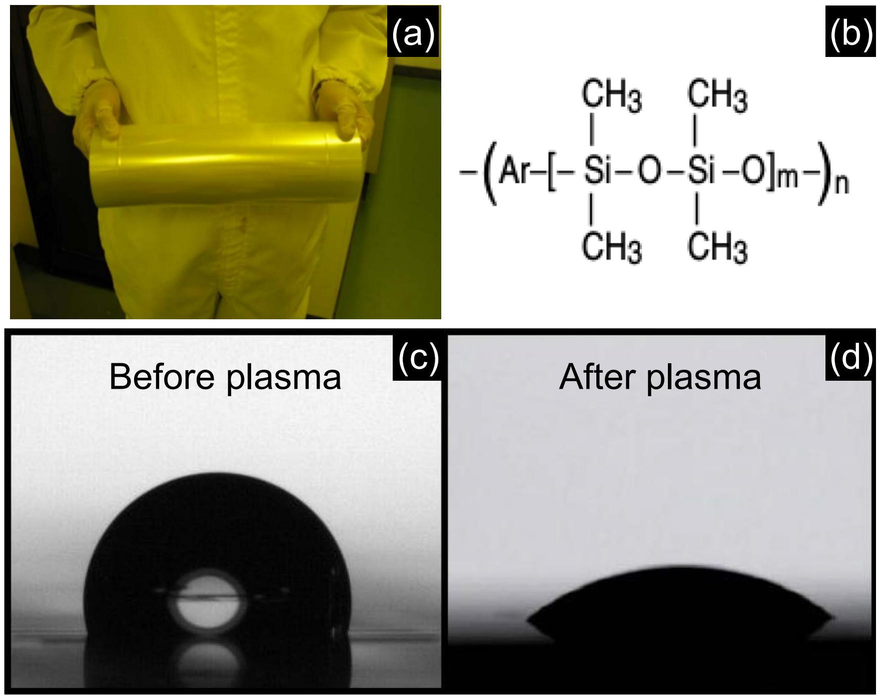
Photo-Patternable Adhesives: SiNR

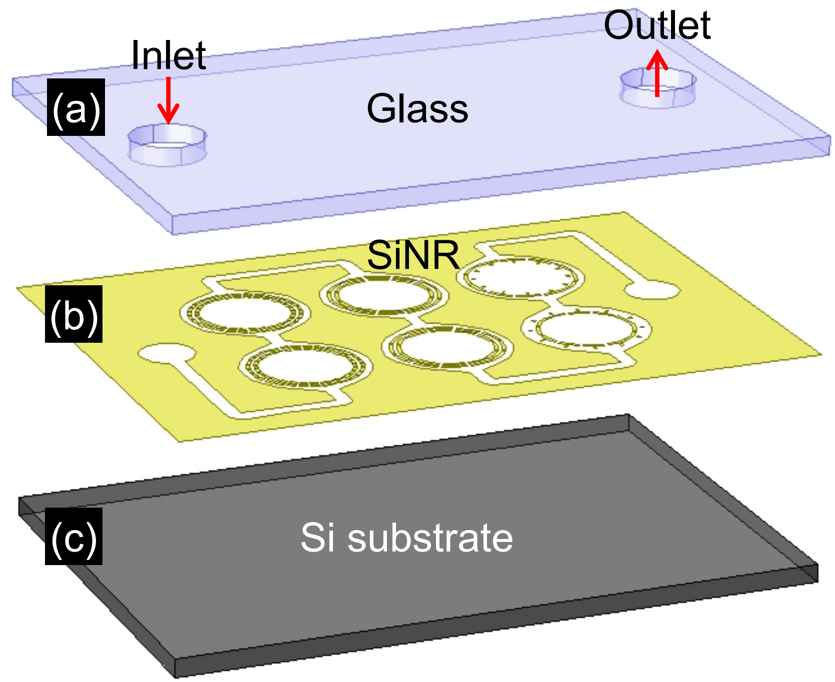
Dry resist for wafer bonding

50µm-thick foil

Dry resist for wafer bonding

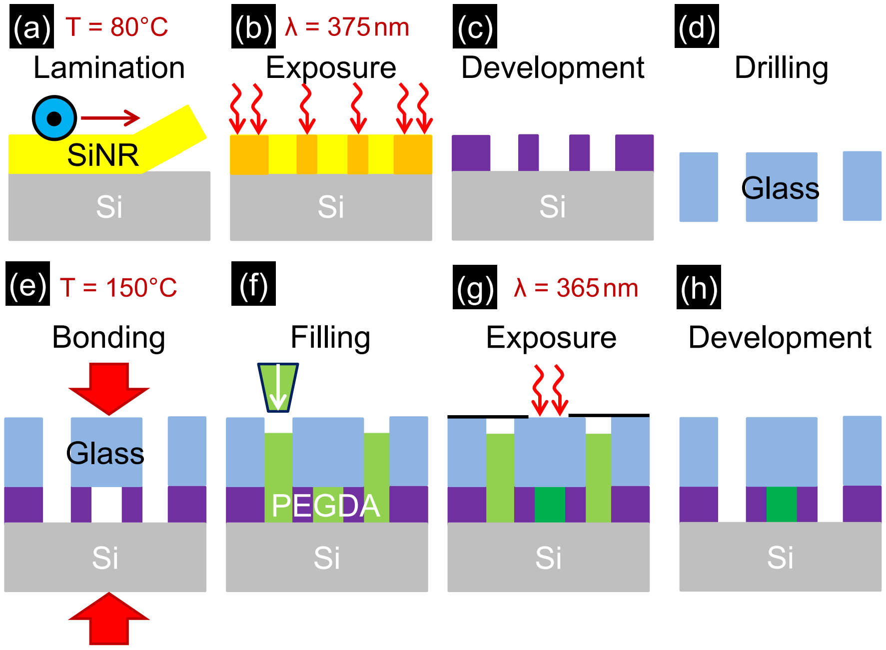
50µm-thick foil

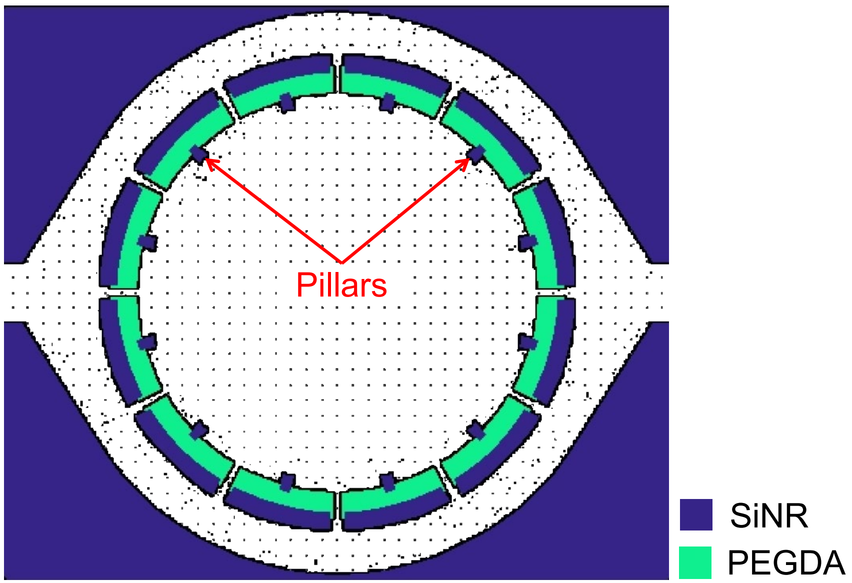


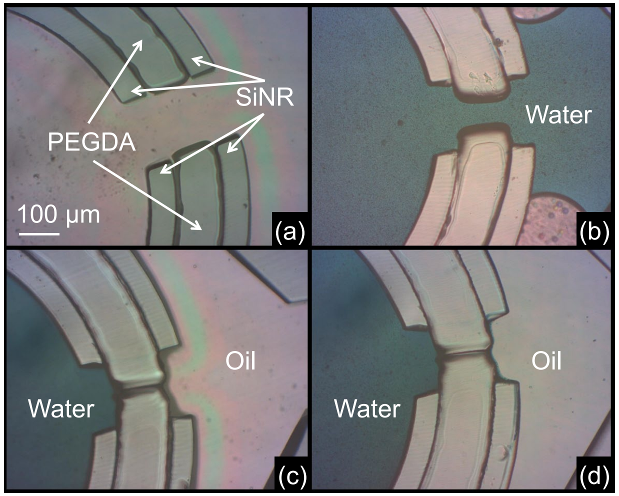


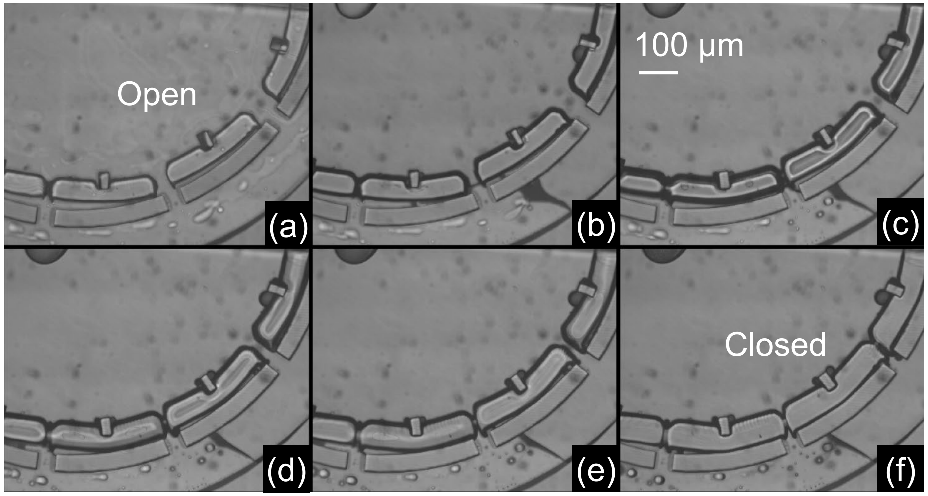
For large-scale manufacturing, better adaptation to industrial processes than PDMS

Example: Hydrogel Self-Sealing Passive Vale

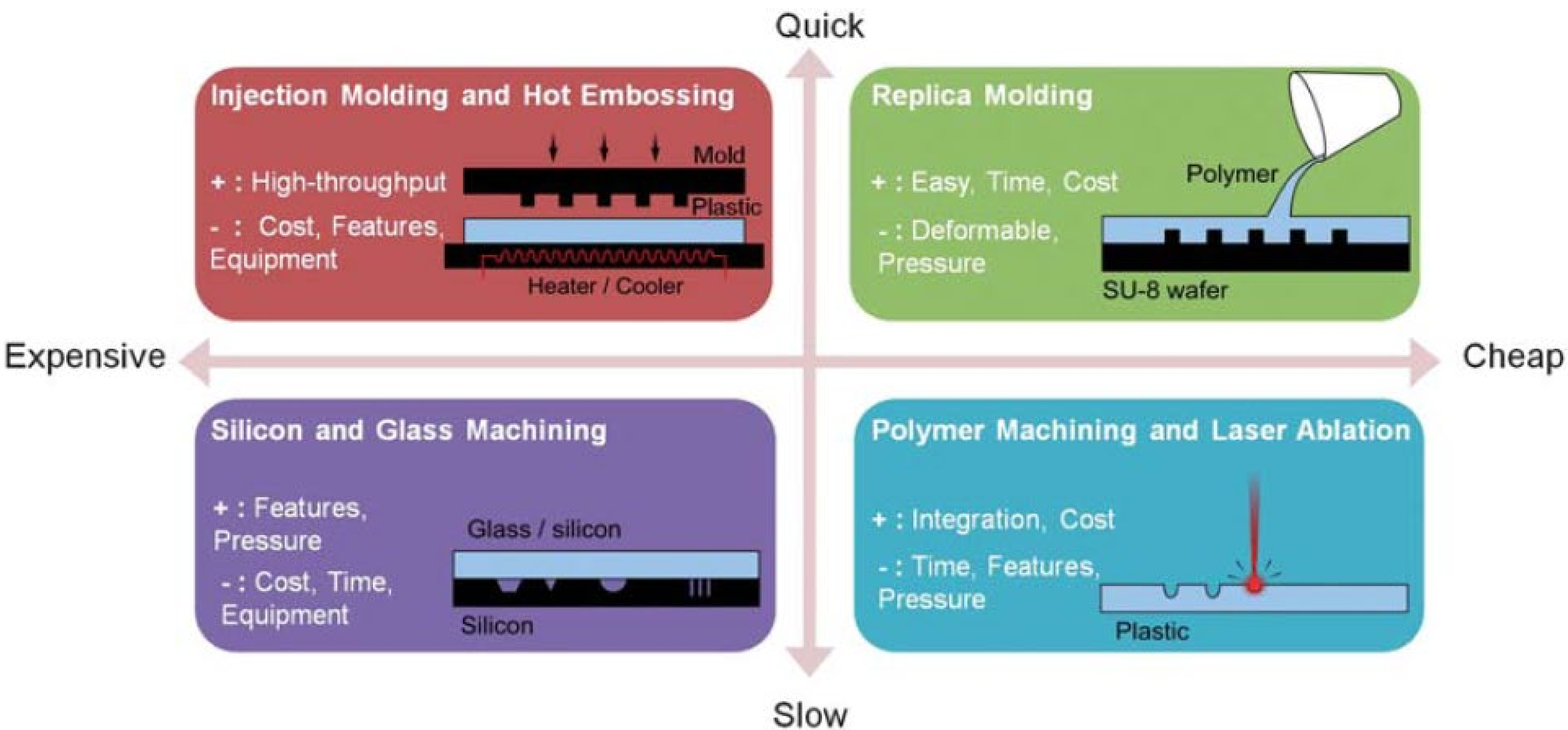








Summary


Quick
Slow
Expensive
Cheap
Injection Molding and Hot Embossing
+ : High-throughput
- : Cost, Features, Equipment
Mold
Plastic
Heater / Cooler
Replica Molding
+ : Easy, Time, Cost
- : Deformable, Pressure
Polymer
SU-8 wafer
Silicon and Glass Machining
+ : Features, Pressure
- : Cost, Time, Equipment
Glass / silicon
Silicon
Polymer Machining and Laser Ablation
+ : Integration, Cost
- : Time, Features, Pressure
Plastic